Joint Spatial–Spectral Optimization for the High-Magnification Fusion of Hyperspectral and Multispectral Images

Jian Long, Yibing Zhan, *Member, IEEE*, Zhengbin Pang, Tong Zhou, Xueqiong Li, Long Lan, and Yuanxi Peng

***Abstract*— The fusion of hyperspectral and multispectral images is an important strategy for enhancing the spatial resolution of hyperspectral images. With the rapid advancement of multispectral imaging technology, the disparity in spatial resolution between multispectral and hyperspectral images is increasing. In certain scenarios, termed high-magnification, this difference can exceed 32×. Previous methods do not perform well under high-magnification fusion, and naturally, a challenge arises in achieving effective high-magnification super-resolution fusion. In light of the above analysis, this article introduces a novel algorithm for high-magnification super-resolution fusion of hyperspectral and multispectral images based on the joint optimization of spatial and spectral information. Specifically, our algorithm consists of three stages: 1) a fast preliminary fusion stage based on the Moore–Penrose inverse and singular value correlation priors for the rapid acquisition of preliminary solutions; 2) a joint spatial–spectral optimization stage where a coupled optimization framework is constructed to achieve integrated optimization of spatial and spectral information; and 3) an error backpropagation optimization stage where an effective error optimization term is introduced to further refine the fusion performance. We conducted extensive experiments on widely employed publicly available simulated datasets and real datasets. The experimental results unequivocally indicate that our proposed methodology consistently exhibits superior fusion performance compared with state-of-the-art methods, even under the condition of ≥60× super-resolution.**

***Index Terms*— High-magnification super-resolution, hyperspectral image super-resolution, image fusion, joint spatial–spectral optimization.**

NOMENCLATURE

LR-HSI	Low-resolution hyperspectral image.
HR-MSI	High-resolution multispectral image.
HR-HSI	High-resolution hyperspectral image.
$\mathbf{Z}$	HR-HSI matrix $\mathbf{Z} \in \mathbb{R}^{L\times W\overline{H}}$.
$\mathbf{H}$	LR-HSI matrix $\mathbf{H} \in \mathbb{R}^{L\times wh}$.
$\mathbf{M}$	HR-MSI matrix $\mathbf{M} \in \mathbb{R}^{l\times W\overline{H}}$.
TSVD	Truncated singular value decomposition.
$\mathbf{U}_*$, $\mathbf{\Sigma}_*$, $\mathbf{V}_*$	Decomposition matrix obtained after TSVD of the * matrix.
$\mathbf{B}$	Spatial fuzzy matrix $\mathbf{B} \in \mathbb{R}^{W\overline{H}\times W\overline{H}}$.
$\mathbf{S}$	Down-sampling matrix $\mathbf{S} \in \mathbb{R}^{W\overline{H}\times wh}$.
$\mathbf{R}$	Spectral response function matrix $\mathbf{R} \in \mathbb{R}^{l\times L}$.
$\mathbf{Z}_1$	Rough estimator of HR-HSI.
$\mathbf{Z}_2$	Preliminary fast fusion result of HR-HSI.
$\mathbf{Z}_3$	Fusion solution after iterative optimization.
μ	$\mu > 0$ is a small real number used to represent the correction coefficient.
$\mathbf{D}$	Spectral matrix.
$\mathbf{C}$	Spatial coefficient matrix.
$\mathcal{M}$	HR-MSI's 3-D tensor $\mathcal{M} \in \mathbb{R}^{W\times\overline{H}\times l}$.
$\mathcal{C}$	Three-dimensional tensor of coefficient matrix $\mathbf{C}$.
$\underline{\mathcal{C}_k}$	kth clustering group of the coefficient tensor $\mathcal{C}$ can be expressed as $\underline{\mathcal{C}_k} \in \mathbb{R}^{p_s\times p_s\times l_s\times K_k}$.
$\mathbf{C}_{\text{nssp}}$	Coefficient matrix that contains nonlocal space self-similarity prior information.
$\|\cdot\|_F$	Frobenius norm.
$\|\cdot\|_{\text{NSSP}}$	Correction term for the coefficient matrix brought by the nonlocal spatial self-similarity prior.
$\mathbf{C}_1$, $\mathbf{\Delta}_1$	Auxiliary variables of the space coefficient matrix C.
$\mathbf{D}_1$, δ_1	Auxiliary variables of the spectral matrix D.
$\|\mathbf{D}\|_{\text{svd}}$	Constraint term for the spectral basis matrix introduced by the singular value decomposition.

Manuscript received 3 April 2024; revised 16 May 2024 and 7 June 2024; accepted 11 June 2024. Date of publication 17 June 2024; date of current version 27 June 2024. This work was supported in part by the Major Science and Technology Innovation 2030 "New Generation Artificial Intelligence" Key Project under Grant 2021ZD0111700; and in part by the Open Fund of the State Key Laboratory of High-Performance Computing, National University of Defense Technology, under Grant 202201-05. *(Corresponding author: Yuanxi Peng.)*

Jian Long, Zhengbin Pang, Tong Zhou, Xueqiong Li, Long Lan, and Yuanxi Peng are with the College of Computer Science, National University of Defense Technology, Changsha 410073, China (e-mail: jian_long@nudt.edu.cn; zbpang@nudt.edu.cn; zhoutong09@nudt.edu.cn; lixueqiong13@nudt.edu.cn; long.lan@nudt.edu.cn; pyx@nudt.edu.cn).

Yibing Zhan is with the JD Explore Academy, Beijing 100000, China (e-mail: zybjy@mail.ustc.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3415121

I. INTRODUCTION

HYPERSPECTRAL imaging technology can obtain mapping information of dozens to hundreds of spectral bands in the same scene. Due to the rich spectral information, hyperspectral images are widely used in remote sensing [1], image processing [2], [3], [4], [5], agriculture [6], mineral surveying [7], the military [8], and other fields. Typically, achieving high spectral resolution results in a loss of spatial

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.

resolution; however, spatial resolution significantly impacts the effectiveness of hyperspectral imagery in various applications. In existing research, to overcome the limitations imposed by insufficient spatial resolution on the application effectiveness of hyperspectral images, there is increasing interest in super-resolution methods that fuse HR-MSIs with LR-HSIs of the same scene. Such fusion methods aim to produce HR-HSIs, offering enhanced spatial details without compromising spectral resolution.

In [9], high-magnification super-resolution refers to the super-resolution reconstruction of conventional images with a magnification of $\geq 32\times$. Similarly, this article introduces the term "high-magnification fusion" to describe fusion super-resolution techniques with a magnification $\geq 32\times$. However, most existing fusion methods have focused mainly on low-magnification super-resolution, such as $8\times$, and research on higher magnification super-resolution, such as $64\times$, has been relatively limited. Nevertheless, many scenes in remote sensing image processing require high-magnification fusion, such as scenes captured by the WorldView-4 satellite, which features a multispectral camera capable of capturing eight bands of multispectral images with a spatial resolution of 1.24 m, and by the GeoEye-1 satellite, which features a multispectral camera that can capture four bands of multispectral images with a spatial resolution of 1.64 m. The panchromatic images captured by these two satellites are used as high-resolution images for fusion, resulting in image spatial resolutions of 0.31 and 0.61 m, respectively. In contrast, satellites equipped with hyperspectral cameras, such as EOS AM-1 and HJ-1A, can capture 36 and 115 bands of hyperspectral images with spatial resolutions as low as 250/500/1000 and 100 m, respectively. Even the most commonly used spaceborne hyperspectral cameras usually have limited spatial resolutions of up to 30 m (multispectral images can achieve super-resolution factors of up to $18\times$–$96\times$) when capturing hyperspectral images. Therefore, researching high-magnification fusion ($\geq 32\times$) is of crucial practical significance. In this article, to emphasize the effectiveness of the proposed method for high-magnification fusion, particular attention is given to fusion scenarios with magnification factors of $\geq 60\times$.

Numerous methods have focused on the fusion of hyperspectral and multispectral images to improve the spatial resolution of hyperspectral images. To further enhance the fusion performance, some studies have utilized strategies incorporating prior information, such as nonlocal spatial self-similarity [10], [11], sparsity [12], [13], [14], [15], nonnegativity [16], and singular value correlation [17]. While these types of prior information enhance the ability to extract either spatial or spectral information from images, in high-magnification fusion scenarios, the significant disparity in pixel quantities between LR-HSIs and HR-MSIs imposes a critical demand for the precise matching and extraction of spatial and spectral information. Nevertheless, these methods often focus on one aspect exclusively (spatial or spectral), overlooking the holistic extraction of spatial–spectral information, thereby limiting overall performance. Specifically, focusing solely on extracting spectral information may lead to the loss of spatial details, while emphasizing only spatial information may confuse spectral features. For instance, Dian and Li [10] introduced a prior on nonlocal spatial self-similarity, effectively enhancing the capability to recover spatial details. However, this approach directly initializes spectral bases from LR-HSIs without further optimization. In high-magnification fusion, the limited number of pixels in LR-HSIs restricts the information available for initializing spectral bases, thereby limiting the overall fusion performance. Moreover, the method proposed in [17] successfully extracts spectral information by leveraging the singular value correlation between LR-HSIs and the target HR-HSI. However, this approach does not specifically optimize spatial information. In high-magnification fusion scenarios, there is a tendency for an imbalance between spatial and spectral information extraction, consequently limiting performance.

Although some methods consider both spectral and spatial information in fusion [18], [19], [20], shortcomings still exist in these methods for high-magnification fusion scenarios. Simoes et al. [18] introduced regularization constraints for optimizing spectral bases and spatial coefficient matrices. However, this method follows a sequential optimization approach, optimizing spectral bases before optimizing coefficient matrices. In scenarios of high-magnification fusion, the limited number of pixels in LR-HSIs, coupled with the inability to provide feedback of subsequently optimized spatial information to the spectral optimization, leads to an imbalance between spectral and spatial information extraction, thereby restricting fusion performance. Zhang et al. [19] and Xie et al. [20] both used deep learning to extract spatial and spectral features simultaneously, but the two methods, like all deep learning methods [21], [22], [23], [24], require a large amount of training data. In the context of high-magnification fusion scenarios, the limited availability of training data leads to poor performance of these methods. In summary, it is still of urgent necessity to develop a method that can efficiently and collaboratively optimize spatial and spectral information within limited data to enhance the super-resolution performance of high-magnification fusion.

Generally, high-magnification super-resolution fusion faces two significant challenges compared to conventional super-resolution fusion. First, the pixel count of the LR-HSI is significantly lower than that of the HR-HSI, making it extremely challenging to extract rich spectral information from the LR-HSI. Second, as the magnification factor increases, the spatial pixel differences between the LR-HSI and HR-MSI rapidly expand. For example, in $32\times$ super-resolution fusion, the difference can reach an order of magnitude of 10^3 (with one LR-HSI pixel corresponding to over 1000 HR-MSI pixels), posing a significant challenge for the joint extraction of spatial–spectral information. In response to these challenges, this article proposes a high-magnification fusion super-resolution algorithm based on joint spatial–spectral optimization. The algorithm consists of three stages: 1) fast fusion for preliminary solutions by utilizing the singular value decomposition of the matrix to acquire an initial fused solution

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.

quickly, and then using this solution, which has the same number of pixels as the target HR-HSI, to initialize the spectral bases, thereby avoiding the limitations of directly initializing the spectral bases from the LR-HSI; 2) joint spatial–spectral optimization by iteratively solving coupled matrix equations with constraints on the nonlocal spatial self-similarity and singular value decomposition correlations, enhancing the collaborative extraction of spatial–spectral features. In addition, by dynamically expanding the dimensions of the spectral bases and coefficient matrices, this approach efficiently utilizes limited information; and 3) error backpropagation optimization, which feeds back the prior knowledge of the LR-HSI through interpolation into the error matrix, further enhancing the fusion performance.

Our main contributions are summarized as follows.

1) We propose a spatial–spectral joint optimization framework for high-magnification fusion that utilizes nonlocal spatial self-similarity and singular value decomposition correlation prior information to extract and integrate spatial–spectral information effectively.
2) We developed two fast fusion strategies based on singular value matrix correlation, providing more accurate initial estimates for spectral bases and spatial coefficient matrices, thereby enhancing the fusion process.
3) By dynamically adjusting the dimensions of spectral bases and coefficient matrices, we further enhance the ability of the joint optimization framework to extract spatial–spectral information.
4) We conducted strict comparisons with 13 state-of-the-art algorithms on two simulated datasets and two real datasets, verifying the superior performance of the proposed method in high-magnification fusion scenarios.

The remainder of this article is organized as follows. Section II introduces related work. Section III provides a detailed description of the proposed method. The experiments and results analysis are outlined in Section IV, and the conclusion is presented in Section V.

II. Related Work

Fusion methods for hyperspectral and multispectral images can be classified into five categories: Bayesian estimation-based methods, spectral unmixing-based methods, matrix factorization-based methods, tensor factorization-based methods, and deep learning-based methods.

Methods based on Bayesian estimation [25], [26], [27], [28] leverage prior information from data statistics to achieve image super-resolution. For instance, in [26], the wavelet domain transform method was used to achieve super-resolution. In [27], the expectation-maximization algorithm was proposed for maximizing the estimation of the fused image. In [28], a spectral dictionary was first learned via Bayesian statistical estimation, followed by the learning of a coefficient dictionary by Bayesian sparse coding. Finally, the spectral dictionary was multiplied by the coefficient dictionary to obtain a fused HR-HSI. However, due to the randomness inherent in probability estimation, the Bayesian method often results in an unstable fusion effect that varies with changes in the random number utilized as a seed.

Fusion methods based on spectral unmixing [29] assume that the image comprises a small number of "pure" spectral features that describe the materials of the scene. Spectral unmixing extracts spectral dictionaries from LR-HSIs and HR-MSIs of the same scene. The target HR-HSI is estimated according to the extracted dictionary and regression coefficient. For instance, in [30], two dictionaries are learned from HSIs and MSIs, and then, the corresponding relationship between the two dictionaries is constructed to obtain the HR-HSI. In addition, Simoes et al. [18] and Bresson and Chan [31] obtained the HR-HSI by transforming the fusion model into a convex optimization equation solution process. However, the quality of dictionary learning significantly influences the quality of fusion by spectral unmixing-based methods.

Fusion methods based on matrix factorization [32] assume that the target HR-HSI can be decomposed into a spectral basis matrix and a coefficient matrix. The target HR-HSI can be obtained in the fusion process by solving for the spectral basis matrix and coefficient matrix from LR-HSIs and HR-MSIs. In [16], an iterative coupled nonnegative matrix factorization process was implemented to optimize the spectral basis and coefficient matrices and achieve a good fusion effect. In addition, researchers have added prior information based on matrix factorization to improve fusion quality. For example, Dong et al. [33] and Akhtar et al. [34] introduced sparsity to exploit the spatial similarity between images. Dian and Li [10] utilized nonlocal spatial self-similarity prior information to enhance fusion quality. Since images can be regarded as matrix data in computer vision image processing, fusion methods based on matrix factorization have excellent advantages.

Methods based on tensor factorization exploit the inherent 3-D spatial characteristics of hyperspectral images. The existing methods based on tensor factorization can be divided into two categories: decomposition methods based on the Tucker decomposition model [35], [36] and decomposition methods based on the canonical polyadic decomposition (CPD) model [37]. The fusion methods based on the Tucker decomposition model assume that the target HR-HSI can be decomposed into the product of a core tensor and three-factor matrices. Consequently, the HR-HSI can be estimated by solving for the core tensor and factor matrices from HR-MSIs and LR-HSIs [35], [36]. However, due to the large amount of computation, these fusion methods take more time. The fusion methods based on CPD assume that the sum of products of 3-D vector groups can represent the HR-HSI. Methods of this type decompose 3-D tensors into 1-D vectors for faster processing [37]. Nevertheless, these methods require the same rank in all three dimensions, which is different from the situation in practice and thus results in unsatisfactory fusion quality.

An increasing number of studies on deep learning-based fusion methods have been conducted in recent years [21], [22], [23], [24], [38], [39], [40], [41], [42]. For instance, in [21] and [22], CNNs were utilized for image fusion, while [23] and [24] proposed a model-inspired network structure to address the fusion problem. In [38], an imaging model was used to initialize the HR-HSI, which was then mapped to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.

the ground-truth image using a trained CNN. In [39], a dual-branch CNN was proposed for fusion that extracts spectral features of each pixel from each LR-HSI and spatial features from each HR-MSI using two separate CNN branches. Despite the relatively good performance of deep learning-based fusion methods, they still have some limitations. Specifically, most of these methods require additional HR-MSI, LR-HSI, and HR-HSI datasets for pretraining, and given that one HSI image usually contains a similar amount of data to hundreds of conventional images, this results in substantial computational costs and limited practical applicability.

III. Proposed Method

As mentioned in Section I, in some application scenarios, the spatial resolution of multispectral images is often significantly greater than that of hyperspectral images, making the exploration of high-magnification fusion techniques particularly important. Therefore, this article innovatively proposes a high-magnification fusion method based on spatial–spectral joint optimization. As shown in Fig. 1, this method consists of three key steps: 1) the fast fusion for preliminary solutions stage, where fast fusion is used to obtain a relatively accurate initial spectral basis $\mathbf{D}$ and coefficient matrix $\mathbf{C}$; 2) the spatial–spectral joint optimization stage, where a coupled optimization framework is used to optimize the spectral basis and coefficient matrix alternately; and 3) the optimization via error backpropagation stage, where the reconstruction error matrix is used to further improve the quality of the fused image. In the following, detailed explanations of each step are presented.

A. Fast Fusion for Preliminary Solutions

In contrast to the traditional approach [10], [43], [44] of directly obtaining initial spectral bases and coefficient matrices from LR-HSI and HR-MSI, the primary objective of the fast fusion for preliminary solutions in this section is to furnish more precise initial spectral bases and coefficient matrices for subsequent joint spatial–spectral optimization.

In this article, we denote the target HR-HSI matrix as $\mathbf{Z} \in \mathbb{R}^{L \times W\overline{H}}$ to indicate that it contains L-bands and $W \times \overline{H}$ pixels. Similarly, the LR-HSI matrix involved in the fusion is denoted as $\mathbf{H} \in \mathbb{R}^{L \times wh}$, indicating that it contains L-bands and $w \times h$ pixels. The HR-MSI involved in the fusion is denoted as $\mathbf{M} \in \mathbb{R}^{l \times W\overline{H}}$ and consists of l bands and $W \times \overline{H}$ pixels. Here, $L > l, W > w$, and $\overline{H} > h$. It is usually assumed that the LR-HSI is the spatial blur downsampling matrix of the target HR-HSI and the HR-MSI is the spectral blur downsampling matrix of the target HR-HSI. Therefore, (1) and (2) can be obtained, where $\mathbf{B} \in \mathbb{R}^{W\overline{H} \times W\overline{H}}$, $\mathbf{S} \in \mathbb{R}^{W\overline{H} \times wh}$, and $\mathbf{R} \in \mathbb{R}^{l \times L}$ represent the spatial fuzzy matrix, downsampling matrix, and spectral response function matrix, respectively. Since various methods [18], [45] can be used in practice to obtain these three matrices, this article assumes that these three matrices are known, and the goal of this article is to solve for the matrix $\mathbf{Z}$

$$\mathbf{H} = \mathbf{ZBS} \tag{1}$$

$$\mathbf{M} = \mathbf{RZ}. \tag{2}$$

In this section, we introduce two fast and efficient approaches for solving for the matrix $\mathbf{Z}$. The first approach is a rapid solution method that combines the Moore–Penrose inverse and singular value decomposition. However, as the Moore–Penrose pseudoinverse is not a true matrix inverse, it introduces errors. Therefore, this approach is best suited for scenarios where the spectral response function is highly accurate, such as in simulated datasets or situations where the spectral response function can be directly obtained from the camera parameters. In situations where the spectral response function needs to be estimated, such as in the experiments conducted on real datasets in this article, the inherent error in the estimated spectral response function may lead to error accumulation when computing the Moore–Penrose pseudoinverse using (3), thus rendering the subsequent singular value decomposition of the computed results infeasible. This article proposes a second method, known as iterative singular value decomposition, to address this issue.

1) Scenario I: In [17], the singular value prior information is used to solve for $\mathbf{Z}$, which is simplified in this article. As shown in (3), first, a rough estimator of the HR-HSI $\mathbf{Z}_1$ is obtained directly using the Moore–Penrose inverse and HR-MSI of the spectral response function

$$\mathbf{Z}_1 = \text{Moore–Penrose}(\mathbf{R})\mathbf{M}. \tag{3}$$

Then, we perform singular value decomposition on the LR-HSI and $\mathbf{Z}_1$ simultaneously and retain the first singular values

$$[\mathbf{U}_h, \mathbf{\Sigma}_h, \mathbf{V}_h] = \text{svds}(\mathbf{H}, \mathbf{1}) \tag{4}$$

$$\left[\mathbf{U}_{z1}, \mathbf{\Sigma}_{z1}, \mathbf{V}_{z1}\right] = \text{svds}(\mathbf{Z}_1, \mathbf{1}) \tag{5}$$

where svds represents the TSVD operation. From the experimental conclusions in [10], we know that the spectral basis of the HR-HSI can be approximated by the matrix $\mathbf{U}$ of the LR-HSI. In [17], it was found that there is a correlation of downsampling factors between the singular values of the LR-HSI and HR-HSI matrices. Therefore, we can obtain a fuzzy solution $\mathbf{Z}_2$ with prior information on the singular value correlation via (6), where df represents the downsampling factor

$$\mathbf{Z}_2 = \mathbf{U}_h(\text{df} \times \mathbf{\Sigma}_h)\mathbf{V}_{z1}^{\mathrm{T}}. \tag{6}$$

Combining (2), we can derive (7). By solving (7) using the method from [17], we can further optimize the fusion values $\mathbf{Z}_2$ using (8). Specifically, we optimize $\mathbf{Z_2}$ by iteratively executing (8) with the iterative termination condition that the value of the root mean square error (RMSE) of all the elements of the matrices $\mathbf{M}$ and $\mathbf{RZ_2}$ no longer decreases

$$\min_{\mathbf{Z}_2} \|\mathbf{M} - \mathbf{RZ}_2\|_F^2 \tag{7}$$

$$\mathbf{Z}_2 = \mathbf{Z}_2 \cdot \left(\mathbf{R}^{\mathrm{T}}\mathbf{M}\right) \cdot / \left(\mathbf{R}^{\mathrm{T}}\mathbf{RZ}_2\right). \tag{8}$$

2) Scenario II: Before introducing the iterative singular value decomposition method, we first present a conjecture: the right singular value matrix obtained from the TSVD by retaining the largest singular value of matrix $\mathbf{Z}$ is similar to the right singular value matrix obtained from the TSVD by retaining the largest singular value of matrix $\mathbf{RZ}$. Similarly,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.

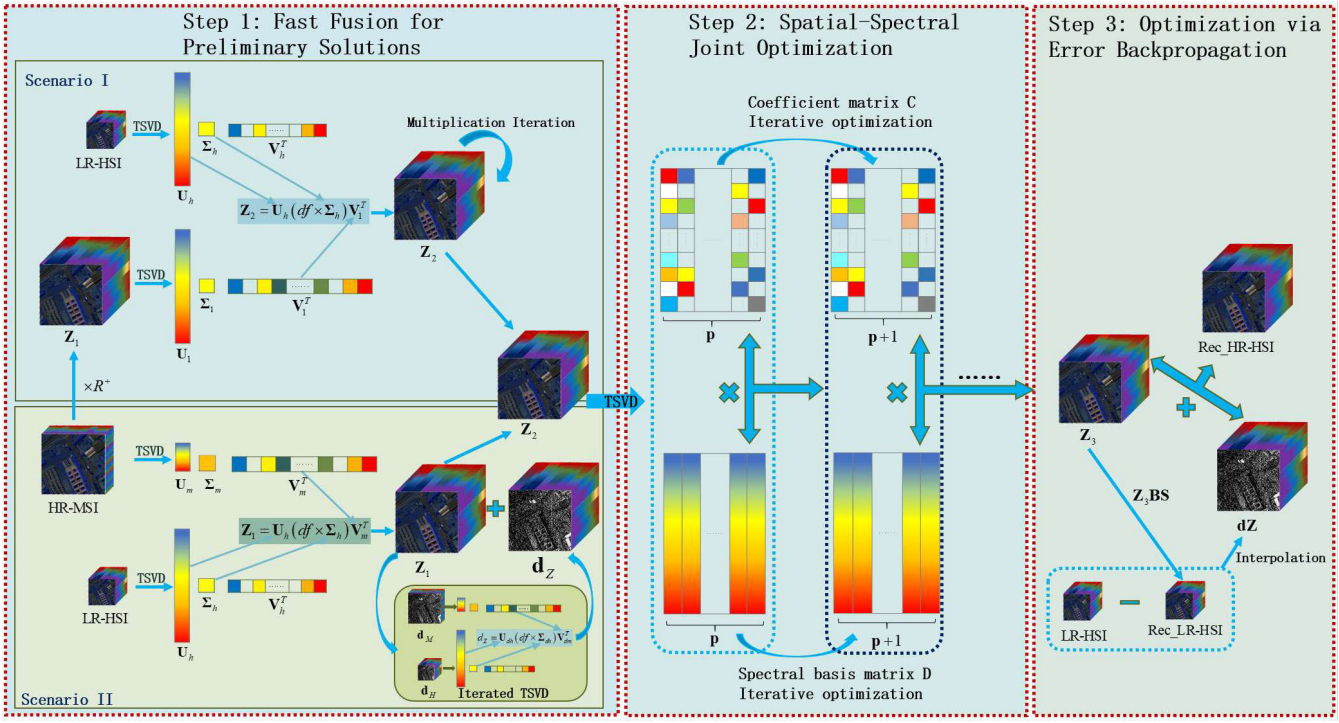


Fig. 1. Proposed high-magnification fusion flowchart. Step 1: a fast fusion solution $\mathbf{Z}_2$ is obtained using two fast fusion schemes based on matrix TSVD correlation. Step 2: the spectral basis matrix D and coefficient matrix C are initialized using TSVD, which retains p singular values, optimizes the complete extraction of the spatial–spectral information via coupled iterations, and increases the feature dimensionality one dimension at a time (in this article, we set the number of iterations to 2). Step 3: the final accurate Rec_HR-HSI is obtained from $\mathbf{Z}_3$ plus the error image $\mathbf{dZ}$ after interpolating the error matrix between Rec_LR-HSI and LR-HSI.

the left singular value matrix is close to the left singular value matrix obtained from the TSVD by retaining the largest singular value of matrix **ZBS**.

Brief Proof: Let $(\mathbf{U}_z, \mathbf{\Sigma}_z, \mathbf{V}_z)$ be the TSVD result for matrix $\mathbf{Z}$ obtained by retaining one singular value and $(\mathbf{U}_r, \mathbf{\Sigma}_r, \mathbf{V}_r)$ be the singular value decomposition result for matrix $\mathbf{R}$ obtained by retaining one singular value. Since the dimensions of $\mathbf{\Sigma}_r$ and $\mathbf{\Sigma}_z$ are 1×1 and the dimensions of $\mathbf{V_r}^{\mathrm{T}} \times \mathbf{U}_z$ are also 1×1, the result of $(\mathbf{\Sigma}_r \times \mathbf{V_r}^{\mathrm{T}} \times \mathbf{U}_z \times \mathbf{\Sigma}_z)$ is a constant. As $\mathbf{U}_r$ and $\mathbf{V}_z$ are truncated singular vector matrices, the TSVD of the matrix $\mathbf{RZ}$ obtained by retaining the largest singular value is approximately equal to $(\mathbf{U}_r, \mathbf{\Sigma}_r \times \mathbf{V_r}^{\mathrm{T}} \times \mathbf{U}_z \times \mathbf{\Sigma}_z, \mathbf{V}_z)$. Similarly, suppose we denote the matrix **BS** as $\mathbf{A}$. In that case, the TSVD results obtained by retaining the largest singular value of matrix $\mathbf{A}$ can be represented as $(\mathbf{U}_a, \mathbf{\Sigma}_a, \mathbf{V}_a)$. Similarly, $(\mathbf{\Sigma}_z \times \mathbf{V_z}^{\mathrm{T}} \times \mathbf{U}_a \times \mathbf{\Sigma}_a)$ is also a constant and can be regarded as a singular value in the TSVD; therefore, the TSVD of the matrix **ZBS** obtained by retaining the largest singular value is also approximately equal to $(\mathbf{U}_z, \mathbf{\Sigma}_z \times \mathbf{V_z}^{\mathrm{T}} \times \mathbf{U}_a \times \mathbf{\Sigma}_a, \mathbf{V}_a)$.

Due to the limitations of TSVD in fully reconstructing the original matrix, the right singular value matrix of $\mathbf{Z}$ is not precisely equal to the right singular value matrix of $\mathbf{RZ}$. Similarly, the left singular matrix of $\mathbf{Z}$ is not precisely equal to that of **ZBS**. However, retaining a TSVD is adequate for recovering most of the information in the original matrix, enabling a rapid initial fusion solution (experimental validation is included in Section IV of this article). According to the above rule, we simultaneously perform TSVD on the hyperspectral image matrix $\mathbf{H}$ and the multispectral image matrix $\mathbf{M}$. We obtain the following results by utilizing the finding in reference [17] that the singular values of the HR-HSI are downsampling multiple of the LR-HSI

$$[\mathbf{U}_h, \mathbf{\Sigma}_h, \mathbf{V}_h] = \mathrm{svds}(\mathbf{H}, \mathbf{1}) \tag{9}$$

$$[\mathbf{U}_m, \mathbf{\Sigma}_m, \mathbf{V}_m] = \mathrm{svds}(\mathbf{M}, \mathbf{1}) \tag{10}$$

$$\mathbf{Z}_1 = \mathbf{U}_h(\mathrm{df} \times \mathbf{\Sigma}_h)\mathbf{V}_m^{\mathrm{T}}. \tag{11}$$

Similarly, if $\mathbf{d}_H = \mathbf{H} - \mathbf{Z}_1\mathbf{BS} = (\mathbf{Z} - \mathbf{Z}_1)\mathbf{BS} = \mathbf{d}_Z\mathbf{BS}$ and $\mathbf{d}_M = \mathbf{R} - \mathbf{RZ}_1 = R(\mathbf{Z} - \mathbf{Z}_1) = \mathbf{Rd}_Z$, then utilizing the above rule enables us to obtain the following results:

$$[\mathbf{U}_{dh}, \mathbf{\Sigma}_{dh}, \mathbf{V}_{dh}] = \mathrm{svds}(\mathbf{d}_H, \mathbf{1}) \tag{12}$$

$$[\mathbf{U}_{dm}, \mathbf{\Sigma}_{dm}, \mathbf{V}_{dm}] = \mathrm{svds}(\mathbf{d}_M, \mathbf{1}) \tag{13}$$

$$\mathbf{d}_Z = \mathbf{U}_{dh}(\mathrm{df} \times \mathbf{\Sigma}_{dh})\mathbf{V}_{dm}^{\mathrm{T}} \tag{14}$$

$$\mathbf{Z}_1 = \mathbf{Z}_1 + \mathbf{d}_Z. \tag{15}$$

As mentioned above, to ensure that the intermediate singular value matrix obtained through TSVD is constant, (12) and (13) continue to utilize the approach of retaining only a single truncated value in the singular value decomposition. In addition, we experimentally demonstrate that iteratively executing (12)–(15) three times can quickly yield a preliminary fast fusion result $\mathbf{Z}_2$.

B. Spatial–Spectral Joint Optimization

To further enhance the fusion quality, we extract spatial and spectral information for optimized images from the inherent

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.

prior information of the images. Therefore, this study proposes an iterative optimization model that couples prior information to obtain a more accurate fusion result. First, we assume that the target HR-HSI can be decomposed into the product of a spectral basis matrix $\mathbf{D}$ and a coefficient matrix $\mathbf{C}$ according to the matrix factorization method, as shown in the following equation:

$$\mathbf{Z} = \mathbf{DC}. \tag{16}$$

Thus, the problem of solving for the HR-HSI is reformulated as finding solutions for a spectral basis matrix and a coefficient matrix. As shown in Algorithm 1, by performing singular value decomposition on matrix $\mathbf{Z}_2$, we can obtain the initial values of the spectral matrix $\mathbf{D}$ and the spatial coefficient matrix $\mathbf{C}$. The dimensions of matrix $\mathbf{D}$ are $\mathbf{D} \in \mathbb{R}^{L\times \mathbf{p}}$, and the dimensions of matrix $\mathbf{C}$ are $\mathbf{C} \in \mathbb{R}^{\mathbf{p}\times W\overline{H}}$. The spectral basis and spatial coefficient matrix are iteratively optimized using the methods described in Sections III-B1 and III-B2.

1) Optimizing the Coefficient Matrix $\mathbf{C}$*:* Since the coefficient matrix mainly contains the spatial information of the fused image and the spatial information mainly comes from the HR-MSI, we can optimize the coefficient matrix $\mathbf{C}$ via

$$\min_{\mathbf{C}} \|\mathbf{M} - \mathbf{RDC}\|_F^2 + \|\mathbf{C}\|_{\text{NSSP}} \tag{17}$$

where $\|\cdot\|_F$ represents the Frobenius norm and $\|\cdot\|_{\text{NSSP}}$ represents the correction term of the coefficient matrix corresponding to the nonlocal spatial self-similarity prior. To solve (17), inspired by the solution in [43], we introduce intermediate variables $\mathbf{C}_1$, initialize $\mathbf{C}_1 = \mathbf{C}$, and construct the function

$$f(\mathbf{C}, \mathbf{C}_1) = \min_{\mathbf{C},\mathbf{C}_1} \|\mathbf{M} - \mathbf{RDC}\|_F^2 + \|\mathbf{C}_1\|_{\text{NSSP}}. \tag{18}$$

We further construct the augmented Lagrangian function (19) of (18)

$$\mathcal{L}(\mathbf{C}, \mathbf{C}_1, \mathbf{\Delta}_1) = f(\mathbf{C}, \mathbf{C}_1) + \mu\|\mathbf{C}_1 - \mathbf{C} - \mathbf{\Delta}_1\|_F^2 \tag{19}$$

where $\mathbf{\Delta}_1$ represents auxiliary variables with the same dimensions as $\mathbf{C}$ and $\mu > 0$ is the minimal real number used to represent the correction coefficient. We solve the augmented Lagrangian function by iteratively solving for the variables one by one.

First, we solve for $\mathbf{C}$

$$\mathbf{C}^{(i+1)} = \arg\min_{\mathbf{C}_1} \|\mathbf{M} - \mathbf{RDC}\|_F^2 + \mu\left\|\mathbf{C}_1^{(i)} - \mathbf{C}^{(i)} - \mathbf{\Delta}_1^{(i)}\right\|_F^2. \tag{20}$$

The solution of (20) can be expressed by (21), where $\mathbf{I}$ is the identity matrix

$$\mathbf{C}^{(i+1)} = \left((\mathbf{RD})^{\mathrm{T}}\mathbf{RD} + \mu\mathbf{I}\right)^{-1} \times \left((\mathbf{RD})^{\mathrm{T}}\mathbf{M} + \mu\left(\mathbf{C}_1^{(i)} - \mathbf{\Delta}_1^{(i)}\right)\right). \tag{21}$$

We solve for $\mathbf{C}_1$ as follows:

$$\mathbf{C}_1^{(i+1)} = \arg\min_{\mathbf{C}_1} \|\mathbf{C}_1\|_{\text{NSSP}} + \mu\left\|\mathbf{C_1}^{(i)} - \mathbf{C}^{(i+1)} - \mathbf{\Delta}_1^{(i)}\right\|_F^2. \tag{22}$$

Inspired by the method of coefficient matrix estimation in [10], we exploit the nonlocal clustering method to extract prior information on the nonlocal space self-similarity of the images. We first convert the HR-MSI to a 3-D tensor $\mathcal{M} \in \mathbb{R}^{W\times\overline{H}\times l}$. Then, $\mathcal{M}$ is divided into $p_s \times p_s \times l$ blocks in the spatial dimension, and the K-means++ clustering algorithm [46] is used to cluster these blocks into K groups. To introduce nonlocal spatial self-similarity prior information in the fusion process, we operate similar to HR-MSI on the coefficient matrix $\mathbf{C}$. The coefficient matrix $\mathbf{C}$ is transformed into a 3-D tensor $\mathcal{C} \in \mathbb{R}^{W\times\overline{H}\times l_s}$, $\mathcal{C}$ is divided into small blocks of dimensions $p_s \times p_s \times l_s$ in the same spatial dimension, and these small blocks are divided into K groups according to the clustering results of $\mathcal{M}$. Here, l_s represents the dimension of the spectral basis. That is, the spectral basis matrix $\mathbf{D}$ can be expressed as $\mathbf{D} \in \mathbb{R}^{L\times l_s}$ (the initial value of l_s is $\mathbf{p}$).

The kth clustering group of the coefficient tensor $\mathcal{C}$ can be expressed as $\underline{\mathcal{C}_k} \in \mathbb{R}^{p_s\times p_s\times l_s\times K_k}$, where K_k represents the number of spatial patches of this clustering group. We convert the 4-D tensor $\underline{\mathcal{C}_k}$ into a 2-D matrix $\underline{\mathbf{C}_k} \in \mathbb{R}^{p_s p_s l_s\times K_k}$ and perform singular value decomposition on the matrix $\underline{\mathbf{C}_k}$

$$\left[\underline{\mathbf{U}_k}, \underline{\mathbf{\Sigma}_k}, \underline{\mathbf{V}_k}\right] = \text{svd}\left(\underline{\mathbf{C}_k}\right). \tag{23}$$

Singular values greater than 0 are retained in $\underline{\mathbf{\Sigma}_k}$, and the diagonal matrix is regenerated. Assuming that there are q_k singular values greater than 0 in $\underline{\mathbf{\Sigma}_k}$, these q_k singular values are used to regenerate the matrix $\underline{\mathbf{C}_k}$ and restore the tensor $\underline{\mathcal{C}_k}$. This process can be represented by

$$\underline{\mathbf{C}_k} = \underline{\mathbf{U}_k}(:, 1:q_k)\underline{\mathbf{\Sigma}_k}(q_k)\underline{\mathbf{V}_k}(:, 1:q_k)^{\mathrm{T}} \tag{24}$$

where $\underline{\mathbf{\Sigma}_k}(q_k)$ represents the diagonal matrix regenerated after retaining singular values greater than 0. For $k = 1, 2, 3, \ldots, K$, the same operation is performed on all the cluster groups of K, and finally, the K cluster groups are restored to the coefficient matrix. After the above operations, the coefficient matrix contains prior information on nonlocal space self-similarity. To distinguish it from the previous coefficient matrix, we denote the matrix after the above operations as $\mathbf{C}_{\text{nssp}}$. Therefore, the following equation gives the solution to the matrix $\mathbf{C}_1$:

$$\mathbf{C}_1^{(i+1)} = \left(\mathbf{C}_{\text{nssp}}^{(i)} + \mu\left(\mathbf{C}^{(i+1)} + \mathbf{\Delta}_1^{(i)}\right)\right)\Big/(1+\mu). \tag{25}$$

Then, we solve for $\mathbf{\Delta}_1$

$$\mathbf{\Delta}_1^{(i+1)} = \mathbf{\Delta}_1^{(i)} - \left(\mathbf{C}_1^{(i+1)} - \mathbf{C}^{(i+1)}\right). \tag{26}$$

2) Optimizing the Spectral Basis Matrix $\mathbf{D}$*:* After obtaining an estimate of the coefficient matrix $\mathbf{C}$, we need to estimate the spectral basis matrix. Similarly, we can obtain the solution for the spectral basis matrix $\mathbf{D}$ by solving

$$\min_{\mathbf{D}} \|\mathbf{H} - \mathbf{DCBS}\|_F^2 + \|\mathbf{D}\|_{\text{svd}} \tag{27}$$

where $\|\mathbf{D}\|_{\text{svd}}$ represents the constraint term on the spectral basis matrix introduced by the singular value decomposition in the coupled optimization. Similarly, to solve (27), we introduce intermediate variables $\mathbf{D}_1$, initialize them as $\mathbf{D}_1 = \mathbf{D}$, and construct function

$$f(\mathbf{D}, \mathbf{D}_1) = \min_{\mathbf{D},\mathbf{D}_1} \|\mathbf{H} - \mathbf{DCBS}\|_F^2 + \|\mathbf{D_1}\|_{\text{svd}}. \tag{28}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.

We solve the function (28) by constructing an augmented Lagrange function

$$\mathcal{L}(\mathbf{D}, \mathbf{D}_1, \boldsymbol{\delta}_1) = f(\mathbf{D}, \mathbf{D}_1) + \mu\|\mathbf{D}_1 - \mathbf{D} - \boldsymbol{\delta}_1\|_F^2 \tag{29}$$

where $\boldsymbol{\delta}_1$ represent auxiliary variables with the same dimensions as $\mathbf{D}$. Similar to solving for the coefficient matrix $\mathbf{C}$, we solve this function by iteratively solving for the variables one by one.

First, we solve for $\mathbf{D}$

$$\mathbf{D}^{(i+1)} = \arg\min_{\mathbf{D}} \|\mathbf{H} - \mathbf{DCBS}\|_F^2 + \mu\left\|\mathbf{D}_1^{(i)} - \mathbf{D}^{(i)} - \boldsymbol{\delta}_1^{(i)}\right\|_F^2. \tag{30}$$

The solution of (30) can be expressed by

$$\mathbf{D}^{(i+1)} = \left((\mathbf{CBS})(\mathbf{CBS})^{\mathrm{T}} + \mu\mathbf{I}\right)^{-1} \times \left(\mathbf{H}(\mathbf{CBS})^{\mathrm{T}} + \mu\left(\mathbf{D}_1^{(i)} - \boldsymbol{\delta}_1^{(i)}\right)\right). \tag{31}$$

To solve for $\mathbf{D}_1$, we define

$$\mathbf{D}_1^{(i+1)} = \arg\min_{\mathbf{D}_1} \|\mathbf{D}_1\|_{\mathrm{svd}} + \mu\left\|\mathbf{D}_1^{(i)} - \mathbf{D}^{(i+1)} - \boldsymbol{\delta}_1^{(i)}\right\|_F^2. \tag{32}$$

The solution for matrix $\mathbf{D}_1$ can be obtained as follows:

$$\mathbf{D}_1^{(i+1)} = \left(\mathbf{D}_{\mathrm{svd}}^{(i)} + \mu\left(\mathbf{D}^{(i+1)} + \boldsymbol{\delta}_1^{(i)}\right)\right)\Big/(1+\mu). \tag{33}$$

Similar to $\mathbf{C}_{\mathrm{nssp}}$, $\mathbf{D}_{\mathrm{svd}}$ represents the solution for imposing a singular value decomposition constraint on the spectral matrix, and we will provide the specific value of this constraint later in the text.

By solving for $\boldsymbol{\delta}_1$, we define

$$\boldsymbol{\delta}_1^{(i+1)} = \boldsymbol{\delta}_1^{(i)} - \left(\mathbf{D}_1^{(i+1)} - \mathbf{D}^{(i+1)}\right). \tag{34}$$

To further improve the optimization effectiveness, we apply the approach introduced in Section III-A2 of this article to further process the spectral basis matrix and coefficient matrix obtained through the process above. First, we solve for the error hyperspectral matrix, $\mathbf{d}_H$, as follows:

$$\mathbf{d}_H = \mathbf{H} - \mathbf{DCBS}. \tag{35}$$

Due to the minimal values of the error multispectral matrix, significant bias may be introduced when performing a TSVD. To address this issue, we use the following equation directly to approximate the solution for $\mathbf{d}_Z$ as:

$$\mathbf{d}_Z = \text{Moore–Penrose}(\mathbf{R})\mathbf{M} - \mathbf{DC}. \tag{36}$$

Performing a TSVD by retaining only the largest singular values of $\mathbf{d}_H$ and $\mathbf{d}_Z$ yields the following equation:

$$\left[\mathbf{d}_f, \ , \ \right] = \text{svds}(\mathbf{d}_H, \mathbf{1}) \tag{37}$$

$$\left[\ , \boldsymbol{\Sigma}_{dz}, \mathbf{V}_{dz}\right] = \text{svds}(\mathbf{d}_Z, \mathbf{1}) \tag{38}$$

$$\mathbf{c}_f = \boldsymbol{\Sigma}_{dz}\mathbf{V}_{dz}^{\mathrm{T}}. \tag{39}$$

In TSVD, the closer the number of retained singular values is to the number of singular values of the original matrix, the higher the reconstruction accuracy between the reconstructed and original matrix. Therefore, we add $\mathbf{d}_f$ to the spectral basis $\mathbf{D}$ by concatenating them along the second dimension (matrix $\mathbf{D}$ first, followed by $\mathbf{d}_f$). This enables us to obtain a new spectral base matrix $\mathbf{D}_1$, and the dimensions of the spectral base are $\mathbf{D}_1 \in \mathbb{R}^{L\times(\mathbf{p}+\mathbf{1})}$. Similarly, we add $\mathbf{c}_f$ above $\mathbf{C}$ along the first dimension to obtain a new spatial coefficient matrix $\mathbf{C}_1$, and the dimensions of the coefficient matrix are $\mathbf{C}_1 \in \mathbb{R}^{(\mathbf{p}+\mathbf{1})\times WH}$.

In this context, the new spectral base matrix $\mathbf{D}_1$ introduces the corresponding constraints of singular value decomposition. Consequently, we can construct $\mathbf{D}_{\mathrm{svd}}$ using (40). Importantly, as i increases, the dimension of the spectral base subspace also increases. Therefore, the dimensions of $\mathbf{D}^{(i+1)}$ match those of $\mathbf{D}_1^{(i)}$ at that particular iteration

$$\mathbf{D}_{\mathrm{svd}} = \mathbf{D}^{(i+1)} + \mu\mathbf{D}_1^{(i)}. \tag{40}$$

To ensure that the dimensions of the error spectral matrix $\boldsymbol{\delta}_1$ and the error coefficient matrix $\boldsymbol{\Delta}_1$ are consistent with those of the spectral matrix and the coefficient matrix, we add a column of all-zero vectors to the error coefficient matrix $\boldsymbol{\delta}_1$ and a row of all-zero vectors to $\boldsymbol{\Delta}_1$.

C. Optimization via Error Backpropagation

Through optimization via error backpropagation, a more accurate solution $\mathbf{Z}_3 = \mathbf{DC}$ can be obtained. By using $\mathbf{Z}_3$, the error for the LR-HSI can be obtained as follows:

$$\mathbf{dH} = \mathbf{H} - (\mathbf{Z}_3\mathbf{BS}). \tag{41}$$

By interpolating $\mathbf{dH}$ to the same dimensions as $\mathbf{Z}_3$, we can obtain $\mathbf{dZ}$

$$\mathbf{dZ} = \text{imresize}(\mathbf{dH}, \mathbf{sd}). \tag{42}$$

Therefore, the accurate fused solution $\mathbf{Z}$ after error backpropagation is

$$\mathbf{Z} = \mathbf{Z}_3 + \mathbf{dZ}. \tag{43}$$

Finally, the iterative optimization fusion method for spatial–spectral joint optimization proposed by us can be described by Algorithm 1.

IV. Experiments

To demonstrate the effectiveness of our proposed method, we conducted comparative experiments with 13 state-of-the-art algorithms on two simulated datasets and two real datasets.

A. Dataset and Evaluation Indicators

1) Dataset: The remote sensing dataset used in this study was the Pavia University dataset [47], which consists of images captured over Pavia, Northern Italy, by a ROSIS sensor. The original Pavia University dataset contains 115 bands (0.43–0.86 μm) with 610 × 340 pixels. After discarding some undesirable bands, we finally chose the image patch with a size of 256 × 256 × 93 pixels in the upper left corner of the image as the reference image. The ground dataset used in this article is the Harvard dataset [48]. This dataset contains 50 indoor & outdoor images under daylight and 27 indoor images under artificial & mixed illumination. Each image contains 31 bands of 1040 × 1392 pixels. Without losing generality, we randomly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Our Algorithm

Require: $\mathbf{H}$ $\mathbf{M}$ $\mathbf{R}$ $\mathbf{B}$ $\mathbf{S}$ $\mathbf{p}$ df $\boldsymbol{\mu}$
  Get $\mathbf{Z_2}$ by equation (8) or III-A2
  Set $[\mathbf{U}_2, \mathbf{\Sigma}_2, \mathbf{V}_2] = \text{svds}(\mathbf{Z}_2, \mathbf{p})$
  Initialize $\mathbf{D} = \mathbf{U}_2$
  Initialize $\mathbf{C} = \mathbf{\Sigma}_2 \mathbf{V}_2^T$
  $\mathbf{C}_1 = \mathbf{C}$
  $\Delta_1 = zeros(size(\mathbf{C}))$
  $\mathbf{D}_1 = \mathbf{D}$
  $\delta_1 = zeros(size(\mathbf{D}))$
  for i=1:2 **do**
    Update $\mathbf{C}$ by equation (21)
    Obtain matrix $\mathbf{C_{nssp}}$ by nonlocal clustering
    Update $\mathbf{C_1}$ by equation (25)
    Update $\mathbf{\Delta_1}$ by equation (26)
    Update $\mathbf{D}$ by equation (31)
    Update $\mathbf{D_{svd}}$ by equation (40)
    Update $\mathbf{D_1}$ by equation (33)
    Update δ_1 by equation (34)
    Update $\mathbf{d_f}$ by equation (37)
    Update $\mathbf{c_f}$ by equation (39)
    Construct a new $\mathbf{D_1}$ using $\mathbf{d_f}$ and $\mathbf{D}$.
    Construct a new $\mathbf{C_1}$ using $\mathbf{c_f}$ and $\mathbf{C}$.
    Expand the dimension of $\mathbf{\Delta_1}$ using an all-zero vector
    Expand the dimension of δ_1 using an all-zero vector
  end for
  $\mathbf{Z}_3 = \mathbf{DC}$
  Update $\mathbf{Z}$ by equation (43)
  return $\mathbf{Z}$

selected an image with a size of 512 × 512 × 31 in the upper left corner of the image as the reference image.

The real dataset used in this study consists of two pairs of data, one from a publicly available remote sensing dataset and the other from a dataset captured by our research group. The LR-HSI in the public dataset has a spectral range of 400–2500 nm, with 220 spectral bands and a spatial resolution of 30 m, and was obtained by the Hyperion sensor on the Earth Observing-1 satellite. The HR-MSI in the public dataset contains 13 bands with a spatial ground resolution of 10 m and was acquired from the Sentinel-2A satellite. After removing the absorption bands, we selected the LR-HSI data obtained by the Hyperion sensor, which have dimensions of 100 × 100 × 89. In addition, we chose a 300 × 300 × 4 image captured by the Sentinel-2A satellite as the HR-MSI. Importantly, it should be noted that the LR-HSI and HR-MSI correspond to the same ground area. The images in the real dataset collected by our research group were captured in a corner of a middle school campus playground in Tianjin utilizing a SPECIM-FX17E hyperspectral camera and an RGB camera. The spectral range of the hyperspectral camera is 0.935–1.72 μm, with a spectral resolution of 3.5 nm. Following registration and alignment processing, the LR-HSI selected in this article were of size 333 × 422 × 224, encompassing 224 spectral bands, while the HR-MSI were RGB images of size 999 × 1266 × 3. Due to the substantial volume of data, for computational convenience, we extracted one band at every five-band interval for each LR-HSI to reconstruct a new LR-HSI image with dimensions of 333 × 422 × 44. To facilitate the validation of the fusion effects on a real dataset and the subsequent verification of the classification metrics, we annotated nine classes extracted from the images. These nine classes were labeled "white_cloth," "Lawn," "Green_shrubs," "Crimson_shrubs," "Red_wall," "Black_cloth," "Soil_ground," "Plastic_track," and "Sand_ground."

2) Evaluation Indicators: We applied the most commonly used image evaluation metrics to evaluate the quality of the fusion. These metrics included the peak signal-to-noise ratio (PSNR) [10], which describes the ratio between the maximum power and the residual error power of the signals of the fused image and the reference image. The larger the PSNR is, the better the fusion quality. The RMSE [10] describes the statistical error of pixel values between a fused image and a reference image, and the smaller the RMSE is, the better the quality of the fusion. The relative dimensionless global error in synthesis (ERGAS) [49] describes the overall quality of the fused image; the smaller the ERGAS is, the better the fusion effect. The spectral angle mapper (SAM) [50] evaluates the spectral distortion of the image after fusion. The smaller the value is, the better the fusion effect. Structural similarity (SSIM) [51], which is the similarity between fused and reference images, is evaluated in terms of brightness, contrast, and structure. The closer the SSIM is to 1, the better the fusion quality. The last evaluation indicator is the time taken for the fusion. The shorter the time is, the better the result is. Furthermore, three evaluation metrics, namely, the overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa), were used to validate the performance of our proposed method on real-world datasets.

B. Comparison Methods

Thirteen state-of-the-art algorithms were selected for comparison in this study, including subspace-based low tensor multirank regularization (LTMR) [10], nonfactorization fusion (NFF) [17], coupled sparse tensor factorization (CSTF) [35], a non-blind version of super-resolution tensor-reconstruction (STEREO) [37], a low tensor-train rank (LTTR)-based method [52], the optimized twin dictionaries (OTDs) method [44], matrix decomposition fusion methods (RLR-MDF and LS-MDF) [43], Hysure [18], CUCaNet [21], MIAE [24], SSR-NET [19], and MHF-net [20]. Among these methods, LTMR [10] is a fusion method that uses prior nonlocal spatial similarity information, NFF [17] is a fusion method that uses prior correlation information of singular values, CSTF [35] and LTTR [52] are fusion methods that are based on the Tucker decomposition model in tensor factorization, and STEREO [37] is a fusion method that is based on the CPD decomposition model in tensor factorization. OTD [44] is a fusion method that uses an OTD (RLR-MDF and LS-MDF) [43], and Hysure [18] is a fusion method that uses the unique properties of matrix decomposition. CUCaNet [21], MIAE [24], SSR-NET [19], and MHF-net [20] are deep learning-based methods. All the source code for the compared methods is publicly available. Excluding CUCaNet [21], MIAE [24], SSR-NET [19], and MHF-net [20], we ran all methods on a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.

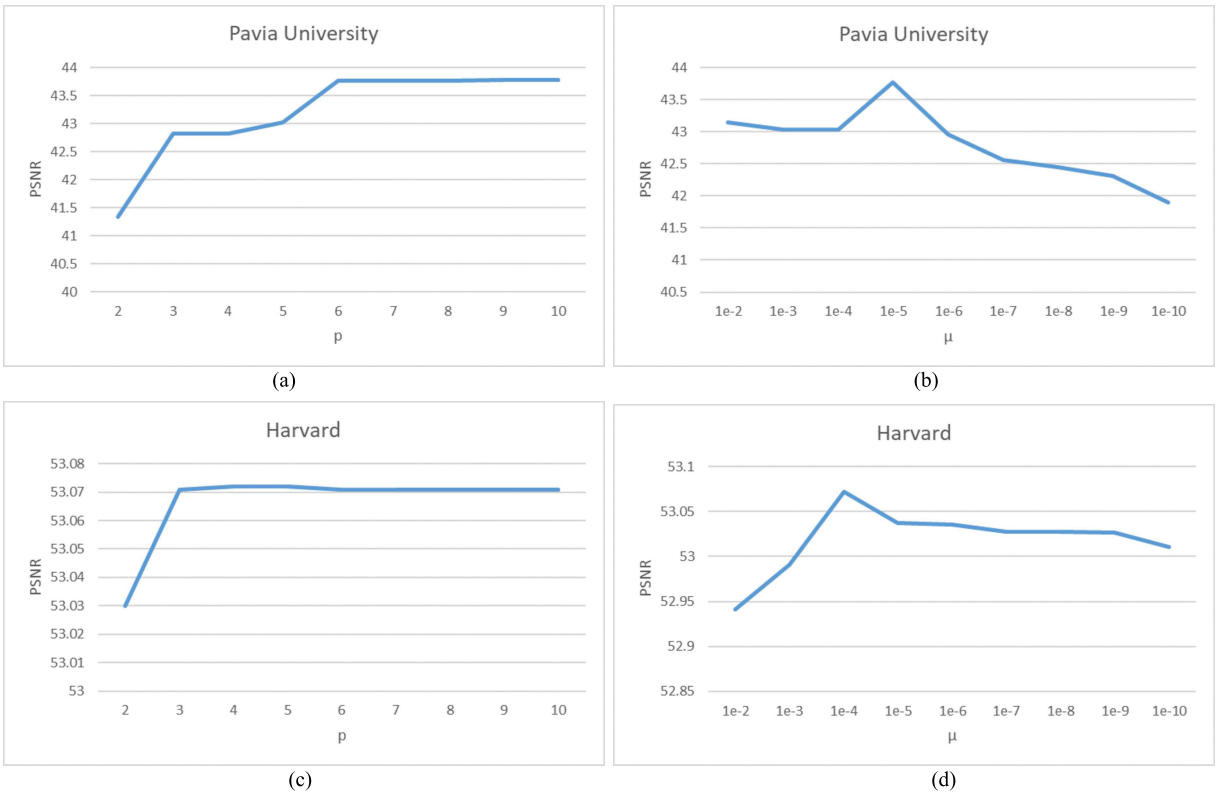


Fig. 2. Diagram for parameter selection. (a) and (b) Pavia University. (c) and (d) Harvard.

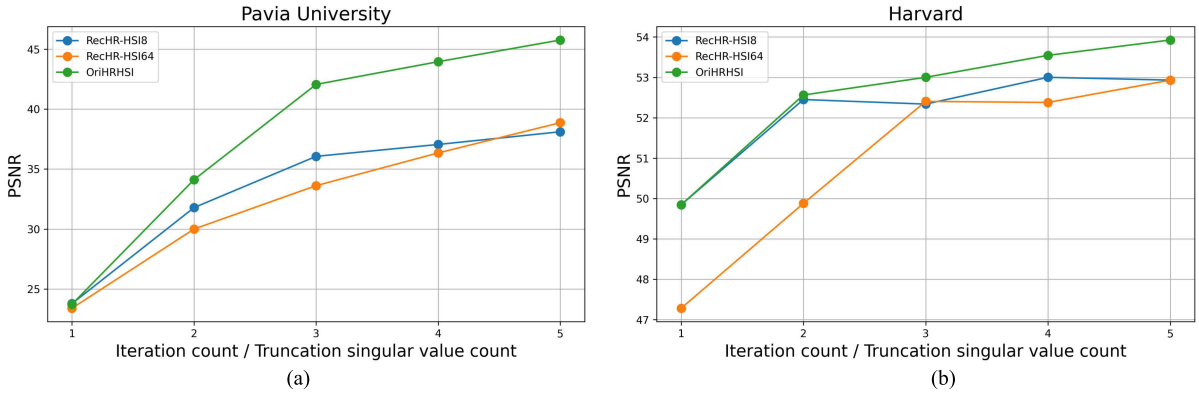


Fig. 3. Iterative singular value decomposition conjecture verification. (a) Pavia University. (b) Harvard.

local host with an Intel[1] Core[2] i7-11700 CPU and 16 GB DDR. CUCaNet [21], MIAE [24], SSR-NET [19], and MHF-net [20] were implemented using the PyTorch platform and trained using 4 NVIDIA TITAN RTX GPUs.

C. Implementation Details

On the Pavia University dataset, we first blurred the reference image with a 7×7 Gaussian filter (with a mean of 2) and then downsampled it to obtain the LR-HSI for fusion. We simulated the HR-MSI by spectrally downsampling the reference image using the IKONOS-class spectral response filter [50]. We used the same Gaussian blur and downsampling method on the Harvard dataset to generate the LR-HSI. In contrast, the HR-MSI was generated using the spectral response function of a Nikon D700 camera. To ensure fair comparison experiments, we adjusted the parameters of all compared methods to achieve optimal fusion performance.

The method proposed in this article primarily addresses the fusion problem arising from high-magnification super-resolution. Therefore, we compare the performances of various methods at $64\times$ super-resolution on both the Pavia University and Harvard datasets to validate the effectiveness of our proposed method using objective evaluation metrics and subjective image displays. In addition, to demonstrate that our method can maintain good fusion performance even at low-magnification super-resolution, we conducted a comparison experiment on both datasets at $8\times$ super-resolution.

[1]Registered trademark.
[2]Trademarked.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.

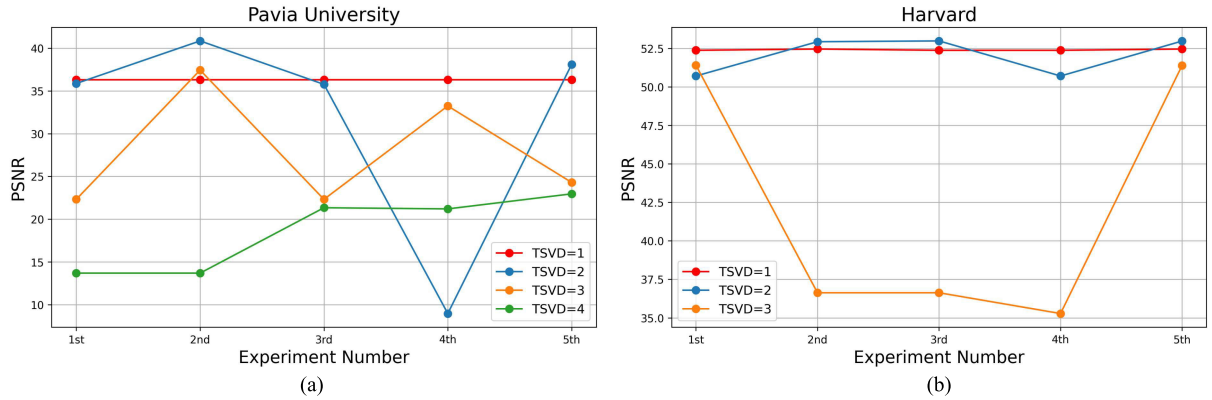


Fig. 4. Comparison of the stability results of fast fusion experiments under different numbers of truncated singular values. (a) Pavia University. (b) Harvard.

Finally, to verify the practicability of the proposed method, we conducted an experimental comparison on a real dataset. The spectral response function estimated by the spectral response function estimation method proposed in [45] was used for all comparison methods on the real datasets. For simplicity and fairness, the point extension functions used by all the comparison methods are the same point spread functions. These are the point spread functions generated by Gaussian blur of size 7×7 and mean 2. It is worth noting that in this article, we used Scenario I for fast fusion for the simulated Pavia University and Harvard datasets because spectral response functions do not need to be estimated. However, since spectral response functions must be estimated for the two real datasets, we used Scenario II for fast fusion.

D. Parameter Selection and Ablation Experiments

1) Parameter Selection: As depicted in Algorithm 1, the method presented in this article centers around two main parameters: $\mathbf{p}$ and μ. $\mathbf{p}$ is utilized to determine the dimension of the spectral basis, impacting the ability to extract spectral information. In general, a larger value of $\mathbf{p}$ can enable the extraction of more spectral information and yield better fusion performance, but this also increases the computation time. The penalty parameter μ is used to correct for the influence of additional variables in the solution formula. To assess the impact of each parameter on the fusion performance, we fixed one parameter and examined how the other parameter affected fusion performance. The observations, illustrated in Fig. 2, reveal the impacts of different parameters on the PSNR values of the fusion results. From Fig. 2(a) and (c), it is evident that $\mathbf{q} = 7$ produces good fusion results on the Pavia dataset, while $\mathbf{q} = 4$ yields the optimal performance on the Harvard dataset. According to Fig. 2(b) and (d), the optimal value can be achieved by setting μ to 1e-5 in the Pavia University dataset and 1e-4 in the Harvard dataset.

2) Ablation Experiments: To validate the conjecture proposed in Section III-A2, comparative experiments were conducted on the Pavia University dataset and the Harvard dataset, as depicted in Fig. 3. The x-axis represents the number of iterations performed using (12)–(15) or the number of retained singular values when applying TSVD to the original HR-HSI matrix. The y-axis represents the PSNR between the reconstructed image and the original image. In the legend, "RecHR-HSI8" refers to the results obtained by applying the proposed regularity at an $8\times$ downsampling rate, "RecHR-HSI64" represents the results obtained at a $64\times$ downsampling rate, and "OriHR-HSI" represents the results obtained by restoring the TSVD of the original HR-HSI matrix. Fig. 3 shows that the results obtained after 1 iteration and the restoration of 1 TSVD from the original HR-HSI matrix are nearly

TABLE I

ABLATION EXPERIMENTS WITH AND WITHOUT (8) IN THE OPTIMIZATION PROCESS

Dataset	Method	PSNR	RMSE	ERGAS	SAM	SSIM	TIME
Pavia/64×	**NoEq.8**	42.829	2.292	0.076	2.341	0.990	4.076
	Ours	**43.769**	**1.981**	**0.064**	**2.011**	**0.991**	**3.913**
Harvard/64×	**NoEq.8**	53.034	0.676	**0.246**	1.740	**0.993**	**6.031**
	Ours	**53.072**	**0.670**	**0.246**	**1.730**	**0.993**	6.286

TABLE II

ABLATION EXPERIMENTS ON THE PAVIA UNIVERSITY DATASET

Pavia University dataset							
df	**Method**	**PSNR**	**RMSE**	**ERGAS**	**SAM**	**SSIM**	**TIME**
8×	**Our1**	40.335	3.405	0.945	3.174	0.985	**1.424**
	Our2	44.278	1.764	0.465	1.733	**0.993**	3.971
	Ours	**44.658**	**1.662**	**0.444**	**1.635**	**0.993**	3.99
64×	**Our1**	40.217	3.502	0.121	3.461	0.985	**1.469**
	Our2	43.749	1.994	**0.064**	2.022	**0.991**	3.887
	Ours	**43.769**	**1.981**	**0.064**	**2.011**	**0.991**	3.913

TABLE III

ABLATION EXPERIMENTS ON THE HARVARD DATASET

Harvard dataset							
df	**Method**	**PSNR**	**RMSE**	**ERGAS**	**SAM**	**SSIM**	**TIME**
8×	**Our1**	52.2	0.78	2.067	1.911	0.992	**1.937**
	Our2	53.037	0.678	1.968	1.747	**0.993**	6.366
	Ours	**53.118**	**0.667**	**1.957**	**1.718**	**0.993**	6.45
64×	**Our1**	52.203	0.78	0.258	1.916	0.992	**1.911**
	Our2	53.042	0.676	0.247	1.745	**0.993**	6.25
	Ours	**53.072**	**0.67**	**0.246**	**1.73**	**0.993**	6.286

TABLE IV

ABLATION EXPERIMENTS ON JOINT SPATIAL–SPECTRAL OPTIMIZATION

Dataset/df	Method	PSNR	RMSE	ERGAS	SAM	SSIM	TIME
Pavia/64×	**Only C**	40.258	3.462	0.119	3.448	0.984	2.941
	Only D	43.632	2.02	0.066	2.066	**0.991**	**3.591**
	Both DC	**43.769**	**1.981**	**0.064**	**2.011**	**0.991**	3.913
Harvard/64×	**Only C**	52.561	0.73	0.255	1.835	**0.993**	4.597
	Only D	53.07	**0.67**	0.247	1.731	**0.993**	**5.241**
	Both DC	**53.072**	**0.67**	**0.246**	**1.73**	**0.993**	6.286

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.

identical. Moreover, as the number of iterations increases, the reconstruction quality improves. However, the increasing loss of information also leads to a larger deviation from the true "OriHR-HSI" result. In addition, the reconstruction results from 8× downsampling are closer to the results obtained by truncating the singular value decomposition of the original matrix. This is because the LR-HSI at 8× downsampling contains more information and can better restore the original matrix. This experiment demonstrates the feasibility of utilizing the conjecture proposed in Section III-A2 to rapidly obtain a fused preliminary solution.

To further validate the significance of retaining only one truncated singular value, as emphasized in Section III-A2, we conducted ablation experiments on different singular value truncation settings in (12) and (13). Considering that multispectral images have a maximum of only four dimensions for the Pavia University dataset, we retained 1–4 truncated singular values on this dataset. Similarly, we retained 1–3 truncated singular values for the Harvard dataset. To ensure the fairness of the experimental results, we repeated the first-stage fast fusion experiment five times under the same experimental conditions. The experimental results are shown in Fig. 4. It is evident from the figure that the results of the five experiments on both datasets remain stable only when the number of truncated singular values is set to 1. However, retaining more than one singular value results in noticeable differences in the results of the five experiments. This indicates that if the process matrices ($\mathbf{\Sigma}_r \times \mathbf{V_r}^{\mathrm{T}} \times \mathbf{U}_z \times \mathbf{\Sigma}_z$) and ($\mathbf{\Sigma}_z \times \mathbf{V_z}^{\mathrm{T}} \times \mathbf{U}_a \times \mathbf{\Sigma}_a$) are not constant, the results show uncertainty. These findings further validate our previous theoretical proof that retaining only one singular value is necessary to ensure the stability and reliability of the fusion results.

In this article, to verify the effectiveness of iterative multiplication optimization for the initial solution using (8) in the fast fusion stage, ablation experiments were conducted under the same experimental conditions with and without the optimization process defined by (8). Table I lists the results of the comparative experiments, where "NoEq.8" indicates the results without optimization based on (8) and "Ours" indicates the results after optimization using (8). The results show that there is an improvement in performance after the optimization process based on (8); this demonstrates the necessity of incorporating the optimization process defined by (8) into the initial solution.

We conducted more ablation experiments to verify the effectiveness of the three components of our proposed algorithm. We defined "Our1" as the fusion effect after only the first step, "Our2" as the fusion effect after the first two steps, and "Ours" as the final effect after completing the entire process. The results of the three processes on the Pavia dataset are shown in Table II, while the effects on the Harvard dataset are presented in Table III. We can observe from both tables that the added modules effectively improve the fusion performance.

To validate the effectiveness of our proposed joint optimization of spatial and spectral information, we defined the fusion method optimized by excluding the spectral matrix D as "Only C." Similarly, the fusion method optimized by excluding the spatial coefficient matrix C was named "Only D." Finally, we referred to the fusion method that incorporates optimization for both spatial and spectral information as "Both DC." Table IV presents the results of ablation experiments on the Pavia and Harvard datasets for 64× super-resolution. The results shown in Table IV demonstrate that the fusion approach, which optimizes both the spectral basis matrix and spatial coefficient matrix simultaneously, outperforms the methods that solely optimize the spectral matrix or spatial coefficient matrix. Since the fusion performance on the Harvard dataset has already reached the limit of the fusion algorithm, the enhancement achieved by simultaneously optimizing spatial–spectral fusion on the Harvard dataset is less pronounced than the improvement observed on the Pavia dataset.

Finally, to validate the importance of dynamically expanding the dimensions of spectral bases and coefficient matrices using the singular value decomposition constraint proposed in this article for enhancing spatial–spectral feature extraction, we conducted an ablation experiment by adding or removing the singular value decomposition constraint in (27). Under the same experimental conditions, we compared the fusion results with and without the constraint to assess its impact. The experimental results are shown in Table V, where "Nosvd" represents the results without constraint optimization, while "Ours" indicates the optimization results after introducing the constraints. The experimental results indicate that introducing constraints can indeed improve fusion performance.

TABLE V

ABLATION EXPERIMENTS WITH AND WITHOUT SPECTRAL BASIS CONSTRAINTS

Dataset	Method	PSNR	RMSE	ERGAS	SAM	SSIM	TIME
Pavia/64×	**Nosvd**	41.115	2.925	0.088	2.508	0.990	**3.867**
	Ours	**43.769**	**1.981**	**0.064**	**2.011**	**0.991**	**3.913**
Harvard/64×	**Nosvd**	52.992	0.679	0.248	1.753	**0.993**	**6.175**
	Ours	**53.072**	**0.670**	**0.246**	**1.730**	**0.993**	**6.286**

TABLE VI

FUSION RESULTS OF THE EXPERIMENTALLY COMPARED METHODS ON THE PAVIA UNIVERSITY DATASET (64×)

Pavia University dataset (64×)						
Method	**PSNR**	**RMSE**	**ERGAS**	**SAM**	**SSIM**	**TIME**
LTMR	40.624	2.596	0.09	2.605	0.987	11.957
NFF	40.232	2.863	0.1	2.898	0.988	1.252
CSTF	36.335	5.031	0.158	4.075	0.974	31.308
STEREO	21.624	21.423	0.763	13.408	0.458	**0.253**
LTTR	32.66	9.368	0.331	8.844	0.943	155.484
OTD	42.598	2.386	0.078	2.451	**0.99**	26.842
LS-MDF	28.883	11.192	0.327	3.508	0.91	**0.48**
RLR-MDF	29.149	10.78	0.316	3.518	0.909	4.704
CUCaNet	38.256	3.763	0.988	3.79	0.984	2533
MIAE	13.692	55.459	14.63	89.864	0.006	358.56
Hysure	**42.725**	**2.305**	**0.076**	**2.354**	**0.99**	26.582
SSR-NET	40.418	2.466	0.722	2.369	0.985	574.8
MHF-net	35.965	4.905	1.4	6.004	0.975	6874.6
Ours	**43.769**	**1.981**	**0.064**	**2.011**	**0.991**	3.913

E. Performance Comparison and Analysis

1) High-Magnification Super-Resolution Results: Table VI presents the fusion results of 13 comparison methods and our method on the Pavia University dataset with 64× super-resolution. We set the parameters of our proposed method according to Section IV-D. The table indicates that our

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.

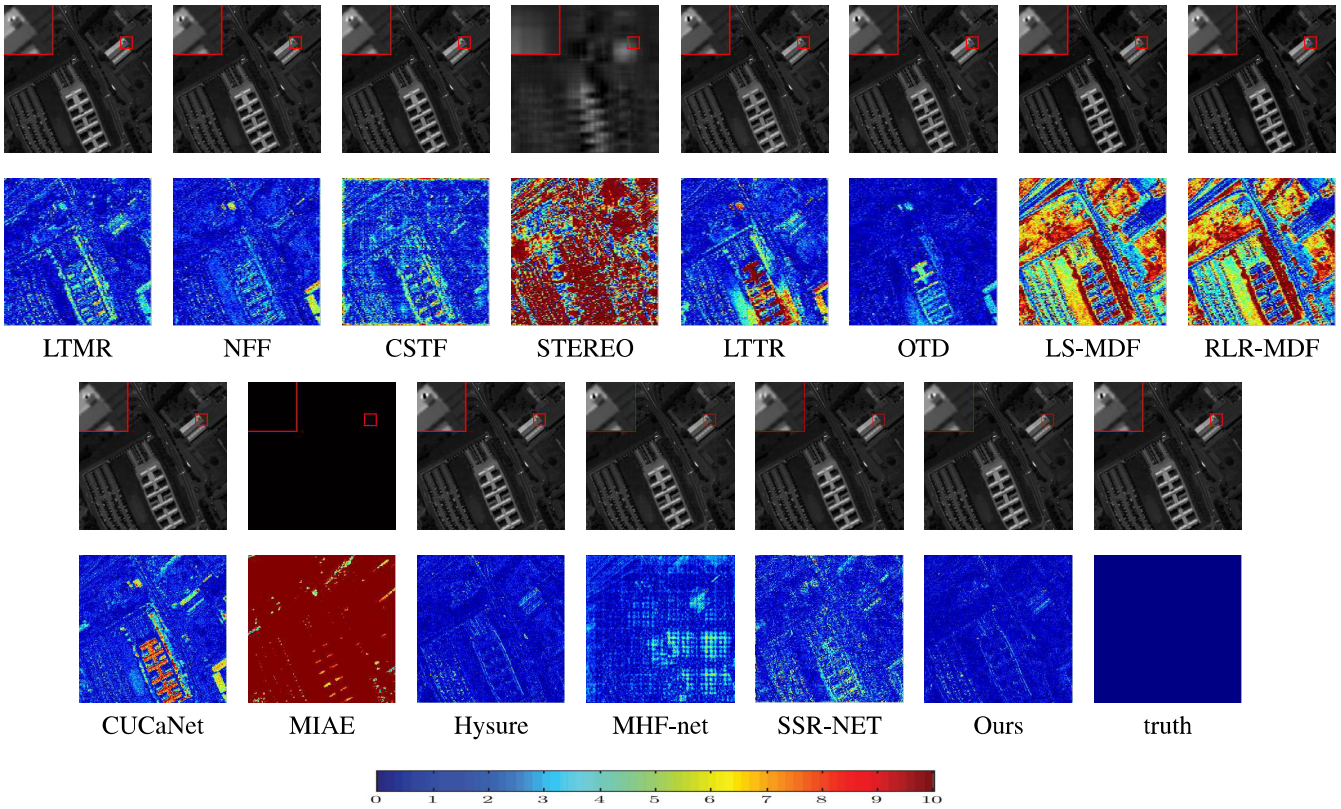


Fig. 5. Fused images of the 20th band obtained by the various comparison methods on the Pavia University dataset. Rows 1 and 3 show the fused images and rows 2 and 4 show the corresponding error images.

proposed method performed best in terms of all objective evaluation indicators. Furthermore, compared to the PSNRs of the optimal results of all the compared methods, our proposed method's PSNR for fusion was improved by more than 1 dB. Apart from our proposed method, the Hysure [18] method achieved the best performance among the remaining methods, but its computation time was almost seven times longer than that of our proposed method. The STEREO [37] method had the shortest fusion time due to its reduction of the data to a 1-D tensor for processing based on CPD. However, since this method requires the rank of each dimension to be the same, its performance was relatively poor in practice because the dimensions were of different ranks. The MIAE [24] method had the worst performance among all the compared methods because it is a data-driven algorithm. When the number of dimensions of the LR-HSIs involved in fusion is minimal, less training data are available, resulting in poor performance.

Fig. 5 shows fused images obtained by all the compared methods on the Pavia University dataset, where the first and third rows show the fused images of the 20th band. The image in the red box in the upper-left corner of the figure is magnified by a factor of four. To visually demonstrate the similarity between each fused image and its ground-truth image, we show the error map between each image and the corresponding ground-truth image in the second and fourth rows. The lower the error is, the closer the image's color is to blue; the larger the error is, the closer the color is to red. As depicted in Fig. 5, our proposed fusion method yields the lowest error.

TABLE VII

FUSION RESULTS OF THE EXPERIMENTALLY COMPARED METHODS ON THE HARVARD DATASET (64×)

Harvard dataset (64×)						
Method	**PSNR**	**RMSE**	**ERGAS**	**SAM**	**SSIM**	**TIME**
LTMR	52.416	0.722	0.265	1.828	**0.992**	4.696
NFF	52.633	0.712	**0.255**	1.826	**0.993**	2.506
CSTF	49.142	1.506	0.309	2.775	0.988	171.631
STEREO	51.113	1.026	0.265	1.964	0.989	**2.131**
LTTR	45.783	2.274	0.468	4.227	0.979	238.207
OTD	48.067	2.018	0.287	2.558	**0.992**	62.826
LS-MDF	47.382	2.257	0.31	1.887	0.987	**0.295**
RLR-MDF	47.738	2.113	0.304	1.992	0.988	14.1329
CUCaNet	45.282	2.743	3.299	5.214	0.982	5039
MIAE	30.602	22.287	12.884	89.515	0.263	1661.7
Hysure	**52.765**	**0.689**	0.257	**1.778**	**0.993**	121.928
SSR-NET	29.729	21.15	61.567	66.05	0.52	957.9
MHF-net	51.354	0.774	3.655	1.884	0.989	10965.77
Ours	**53.072**	**0.67**	**0.246**	**1.73**	**0.993**	6.286

Table VII presents the comparison results of 13 methods, including our proposed method, on the Harvard dataset. Our method outperformed all the other approaches in terms of all objective evaluation metrics. Among the other compared algorithms, Hysure [18] achieved the best performance. Our algorithm improves approximately 0.3 dB in fused PSNR, and its fusion time is only 6.286 s compared to the Hysure [18] algorithm. In accordance with the observations on the Pavia University dataset, deep learning-based methods tended to perform poorly due to their data-driven nature compared to other algorithms on the Harvard dataset. In particular, the SSR-NET method requires at least 12 images for training in its original setup. However, to make a fair comparison with other

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.

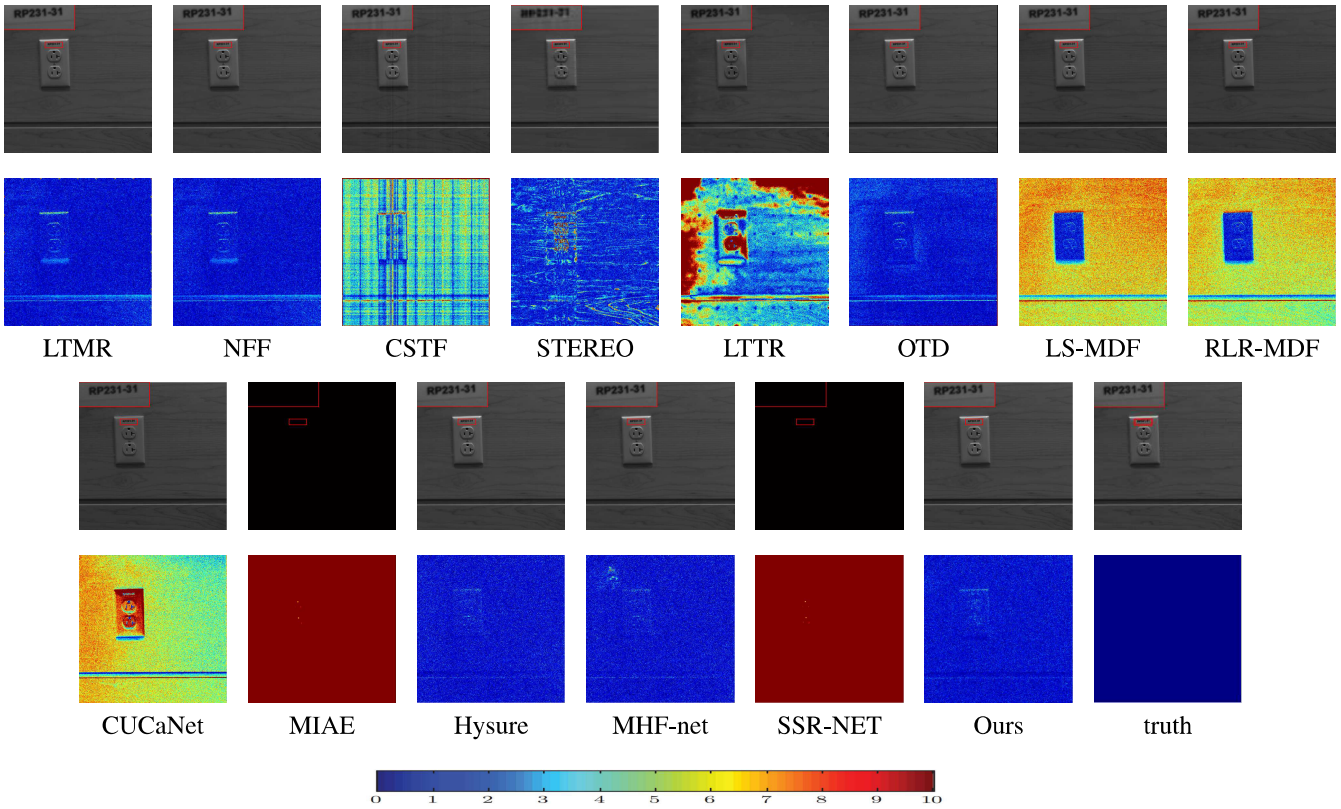


Fig. 6. Fused images of the 20th band obtained by the various comparison methods on the Harvard dataset. Rows 1 and 3 show the fused images and rows 2 and 4 show the corresponding error images.

TABLE VIII

FUSION RESULTS OF THE EXPERIMENTALLY COMPARED METHODS ON THE PAVIA UNIVERSITY DATASET (8×)

Pavia University dataset (8×)						
Method	**PSNR**	**RMSE**	**ERGAS**	**SAM**	**SSIM**	**TIME**
LTMR	41.325	2.573	0.643	1.985	0.991	5.903
NFF	42.307	2.279	0.61	2.3	0.99	**1.246**
CSTF	39.508	3.009	0.765	2.222	0.989	72.636
STEREO	39.974	3.137	0.815	3.578	0.962	13.74
LTTR	40.277	3.03	0.797	2.445	0.986	153.413
OTD	**44.765**	1.699	0.452	1.747	0.992	21.844
LS-MDF	41.829	2.384	0.601	1.947	0.99	**1.036**
RLR-MDF	41.848	2.232	0.586	1.861	0.99	12.925
CUCaNet	**44.683**	**1.56**	**0.433**	**1.416**	**0.994**	3481
MIAE	44.344	1.765	0.471	1.777	0.992	443.5
Hysure	43.945	1.92	0.499	2.013	0.991	25.811
SSR-NET	40.584	2.413	0.707	2.301	0.985	631.8
MHF-net	44.602	1.705	0.448	1.756	**0.993**	4955
Ours	44.658	**1.662**	**0.444**	**1.635**	**0.993**	3.99

TABLE IX

FUSION RESULTS OF THE EXPERIMENTALLY COMPARED METHODS ON THE HARVARD DATASET (8×)

Harvard dataset (8×)						
Methods	**PSNR**	**RMSE**	**ERGAS**	**SAM**	**SSIM**	**TIME**
LTMR	52.095	0.789	2.113	1.819	**0.992**	4.828
NFF	52.72	0.703	2.028	1.809	**0.993**	2.577
CSTF	51.084	1.028	2.145	1.891	0.991	173.941
STEREO	50.692	1.113	2.213	1.995	**0.992**	243.934
LTTR	51.515	0.984	2.045	1.868	0.989	**2.489**
OTD	48.383	1.987	2.227	2.385	**0.993**	79.214
LS-MDF	52.301	0.752	2.103	1.817	**0.992**	**0.3451**
RLR-MDF	51.98	0.8	2.106	1.934	**0.992**	12.496
CUCaNet	53.093	0.671	**1.951**	1.721	**0.993**	5185
MIAE	**53.196**	**0.658**	**1.947**	**1.7**	**0.993**	1869.7
Hysure	53.029	0.674	1.969	1.747	**0.993**	124.358
SSR-NET	29.833	21.191	62.951	66.023	0.519	990.1
MHF-net	52.962	0.668	2.093	1.726	**0.993**	6367.4
Ours	**53.118**	**0.667**	1.957	**1.718**	**0.993**	6.45

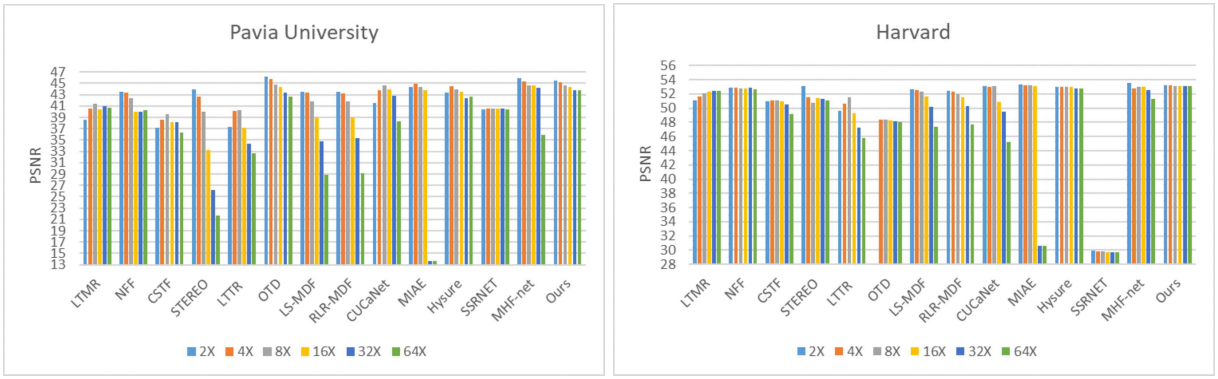
methods, we used only one image for training and testing in this section, leading to a significant decrease in performance.

Similarly, Fig. 6 shows the fused images obtained via the comparison methods on the Harvard dataset. It is evident from the error maps of the fused images that our proposed methods have minimal errors in their fused error maps. Their error maps are almost entirely blue, indicating that the fused images are almost identical to the ground-truth images.

2) Conventional Super-Resolution Results: To demonstrate the efficacy of our approach across traditional super-resolution scales, we conducted a comparison experiment targeting an 8× super-resolution scenario. Table VIII presents the experimental results of various methodologies on the Pavia University dataset. The CUCaNet method exhibited superior performance among all the compared algorithms, but it required 3481 s for execution. In contrast, our proposed method achieved nearly identical performance to CUCaNet and executed in only 3.99 s, representing a mere 0.1% of CUCaNet's runtime. Naturally, our approach offers superior cost-effectiveness.

Table IX shows the experimental results of the various comparison methods on the Harvard dataset. According to the table, the deep learning-based methods (except SSR-NET) performed better overall than the other types of methods. Our proposed approach demonstrated the second-best performance among all the methods. Notably, in terms of some

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 7. PSNRs of the comparison methods at different super-resolution magnifications. (a) Pavia University. (b) Harvard.

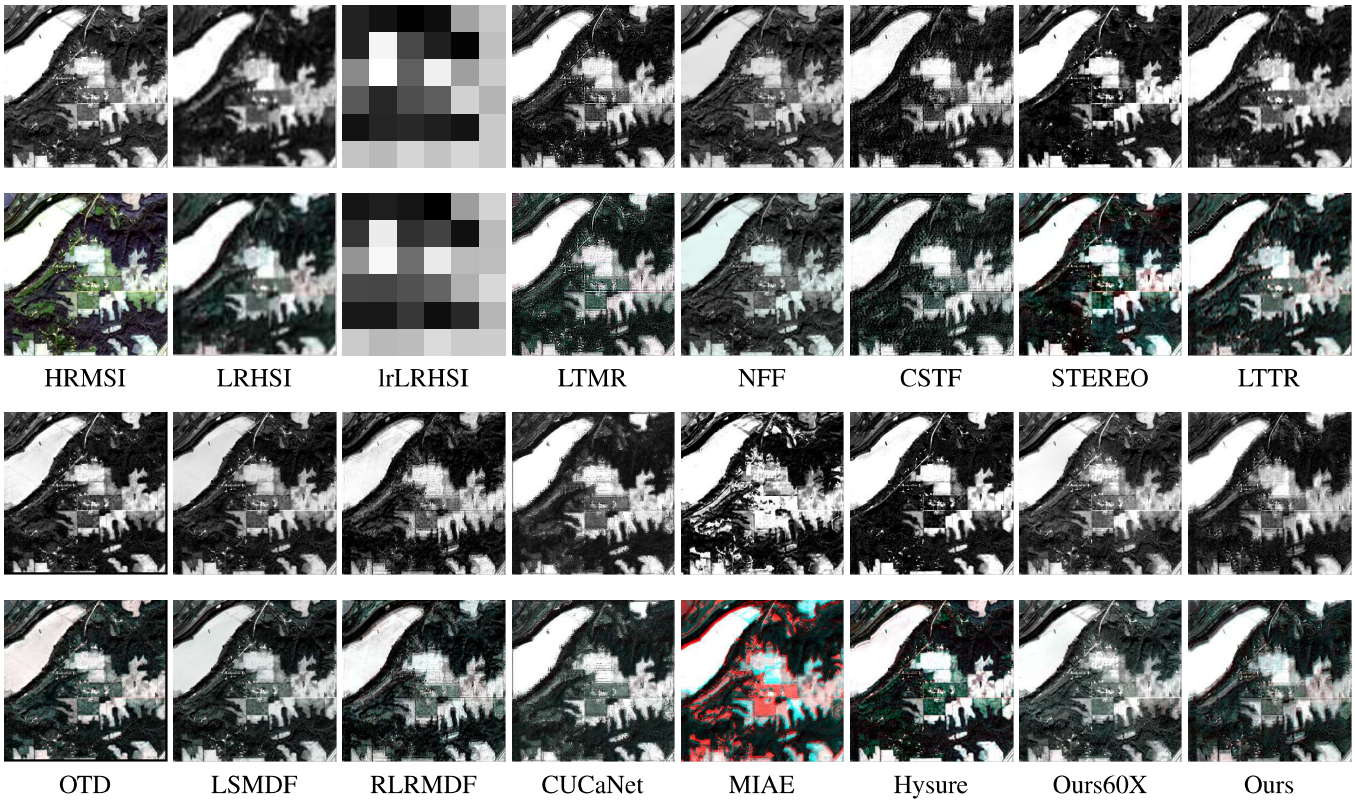


Fig. 8. Fused images obtained by the comparison methods on the first real dataset. The first and third rows show images of the first band of the resulting images, and the second and fourth rows show false-color images of the first, second, and third bands of the resulting images.

metrics, such as SSIM, state-of-the-art performance levels were attained. Compared with the optimal MIAE method, our proposed method had a maximum difference in PSNR of only 0.076 dB, and the differences in all other metrics were less than 0.02, which is almost negligible. However, while the MIAE method took 1869 s to run on a server with better hardware performance, our proposed method took only 6.45 s, which was only approximately 0.3% of the time taken by the MIAE method. Overall, our proposed method's performance was optimal, fully demonstrating its effectiveness.

To further confirm the superior performance of our proposed method, Fig. 7 shows a histogram of the PSNR values of the fused images for all methods at different super-resolution magnifications (note that when performing $2\times$ super-resolution on the Harvard dataset, the OTD method was unable to perform the necessary fuzzy matrix singular value decomposition operations due to memory constraints, so there is no fusion solution). As depicted in the figure, our proposed approach maintains superior fusion performance at various super-resolution levels. Specifically, only the OTD method performed similar to our proposed method on the Pavia University dataset. However, in the $64\times$ super-resolution case, our proposed method outperformed OTD by 1 dB. In addition, OTD took 26.842 s, while our proposed method took only 3.913 s, which was approximately $6.8\times$ faster than OTD. On the Harvard dataset, only the Hysure method maintained a trend similar to that of our proposed method. However,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.

TABLE X
OBJECTIVE METRICS FOR IMAGE CLASSIFICATION UNDER VARIOUS FUSION ALGORITHMS

		LTMR	NFF	CSTF	STEREO	LS-MDF	RLR-MDF	MIAE	Hysure	MSI	Ours	Ours60X
SSFTT	**Kappa(%)**	91.82	95.08	92.74	**98.99**	95.15	98.45	98.23	97.80	95.27	**98.66**	**99.53**
	OA(%)	93.73	96.21	94.38	**99.22**	96.27	98.80	98.63	98.30	96.36	**98.96**	**99.63**
	AA(%)	90.42	94.97	92.64	**98.27**	94.97	97.98	97.92	97.42	94.10	**98.57**	**99.68**
SSRNet	**Kappa(%)**	94.83	96.30	97.61	**98.55**	97.11	97.95	97.99	98.08	97.40	**98.28**	**99.23**
	OA(%)	96.01	97.12	98.14	**98.88**	97.77	98.42	98.45	98.51	97.99	**98.66**	**99.40**
	AA(%)	94.08	96.18	97.80	**98.47**	95.98	96.98	97.18	97.22	96.23	**98.49**	**99.56**
HybridSN	**Kappa(%)**	92.98	93.90	97.48	**98.33**	97.24	97.08	97.43	97.24	-	**97.56**	**99.35**
	OA(%)	94.61	95.31	98.05	**98.71**	97.87	97.73	98.02	97.87	-	**98.11**	**99.50**
	AA(%)	90.06	86.85	96.29	**97.91**	96.33	96.36	96.34	95.67	-	**96.49**	**99.28**

the performance of our proposed method was approximately 0.3 dB greater than that of Hysure in the 64× super-resolution case. In addition, Hysure took 121.928 s, while our method took only 6.286 s, which was approximately 19.3× faster than Hysure.

3) Results on Real Datasets: To verify the practicability of our proposed method, we conducted comparative experiments on two pairs of real datasets. Note that we do not include the SSR-NET or MHF-net method results in this section due to their dependence on HR-HSI data for training, which was not available in the real dataset. To verify the effectiveness of our proposed method in achieving high-magnification super-resolution fusion on real datasets, we performed 20× blur downsampling (using a 7 × 7 Gaussian filter with a mean square of 2) on the LR-HSI dataset obtained from the real dataset. This resulted in a lower resolution lrLR-HSI. We then fused the lrLR-HSI image with the HR-MSI image from the real dataset and recorded the fused result as Ours (60X). Notably, the size of the lrLR-HSI obtained after blur downsampling on the publicly available real dataset was 5 × 5 × 89, whereas the size of the lrLR-HSI obtained after blur downsampling on our own captured real dataset was 17 × 22 × 44. For the parameter settings, we refer to the experimental rule from Fig. 2 that the more spectral bands there are, the larger p is (e.g., the Pavia University dataset), and μ between 1e-4 and 1e-5 is optimal. Therefore, our parameter settings on the real dataset were "$p = 7$ and $\mu = 1e-4$."

Fig. 8 intuitively shows the fusion results of our proposed method and the comparison methods on a publicly available real dataset. The first and third rows show the image of the first band of the resulting image, and the second and fourth rows show the false-color image of the composition of the resulting image's first, second, and third bands. Ours (60X) achieved 60× super-resolution by fusing the lrLR-HSI and HR-MSI, whereas the other methods realized 3× super-resolution by fusing the LR-HSI and HR-MSI. An examination of the third row of images confirms that the fused images produced by both Ours and Ours (60X) closely match the details of the HR-MSI, validating the resolution improvement in the fused images. In addition, analysis of the fourth row reveals close color similarity between our fused false-color image and the LR-HSI false-color image, further confirming the restoration of superior spectral properties. Furthermore, despite the initial blurriness of the lrLR-HSI, our method achieved a comparable 3× super-resolution effect when fusing it with the HR-MSI, demonstrating its efficacy on publicly available real datasets.

According to Fig. 8, most of the methods, including STEREO and Hysure, produced image fusion visualization results similar to those of our proposed method. To further quantitatively verify the superior performance of our proposed method, we captured a pair of real datasets that can be used for classification and performed classification comparison experiments on the fused images. The OTD, CUCaNet, and LTTR methods were unable to complete the fusion operation on this dataset due to the large spatial dimensions of the dataset, which caused the OTD and CUCaNet algorithm computations to exceed the memory limitations, as well as the inconsistency of the spatial dimensions in terms of length and width, which prevented the LTTR algorithm from learning the dictionary. In addition, other methods faced difficulties in fusing the lrLR-HSI with the HR-MSI due to the mismatch in spatial sizes between the lrLR-HSI (17 × 22 × 44) obtained from LR-HSI 20× downsampling and the HR-MSI (999 × 1266 × 3). However, our proposed method addresses this issue by exclusively extracting spectral bases from hyperspectral images and determining the spatial size using spatial coefficients derived from multispectral images.

Fig. 9 shows the fusion results of various methods on the real dataset that we used. Similar to Fig. 8, the first and third rows show the first band of the generated image, and the second and fourth rows show false-color images consisting of the generated image's first, second, and third bands. Similarly, as shown in the figure, the image fused using our proposed method exhibits spectral properties similar to those of the LR-HSI while maintaining spatial properties akin to those of the HR-MSI. Notably, the fusion results of the STEREO method exhibit significant discrepancies with those of the LR-HSI method, resulting in visually inferior outcomes.

To provide an objective assessment of the effectiveness of various fusion algorithms on real datasets, we utilize the fused images resulting from each algorithm's application to our captured data as input for a classification task. Subsequently, we employ classification performance metrics to evaluate the quality of the fusion. In this article, three open-source classification algorithms are used for classification. The SSFTT [53] method utilizes 1% of the data for training and 99% for testing, with the epoch set to 10, while the rest of the parameter settings remain consistent with those in the original article. The SSRNet [54] and HybridSN [55] methods utilized 0.5% of the data for training, another 0.5% for validation, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.

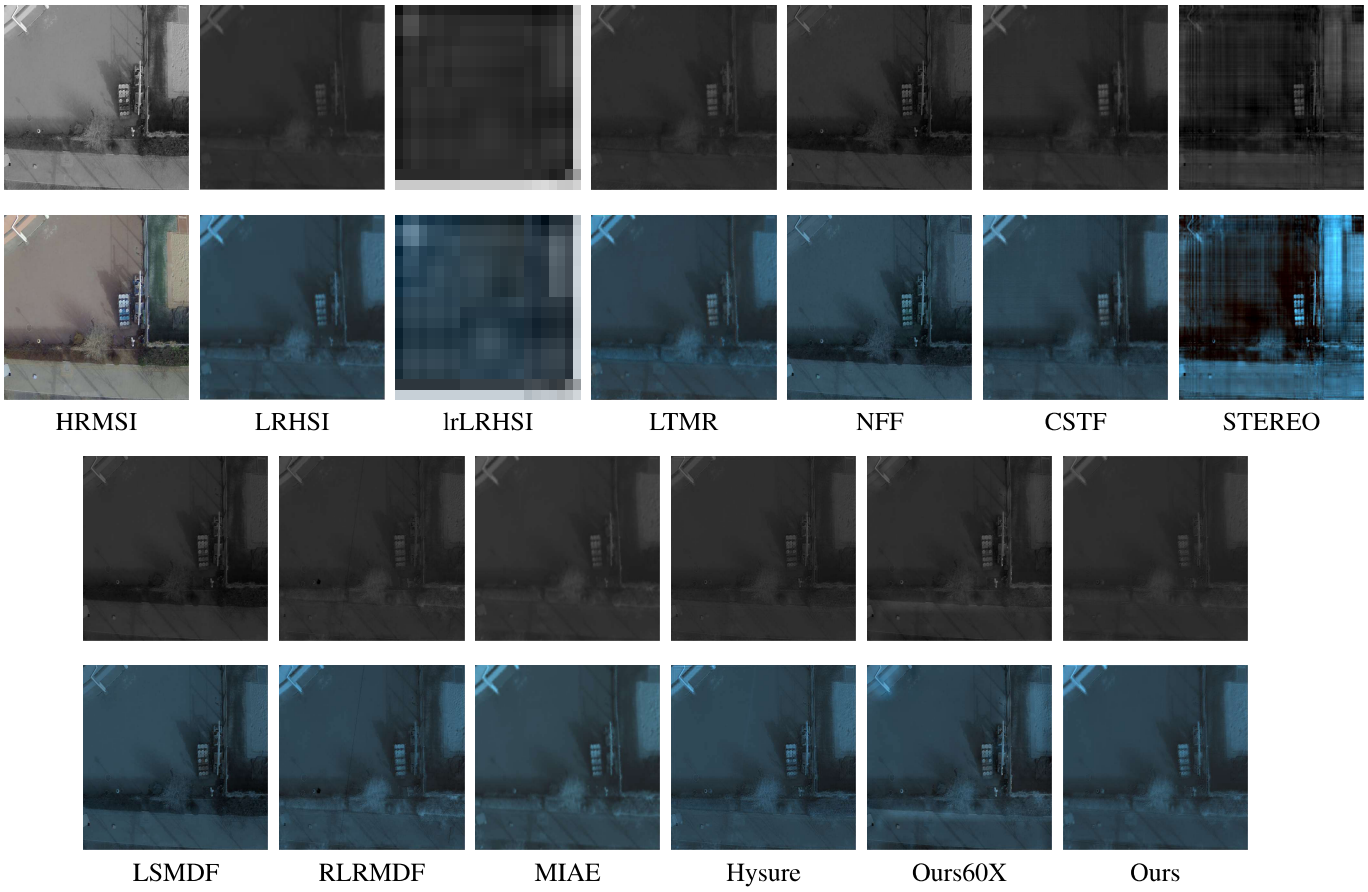


Fig. 9. Fused images obtained by the comparison methods on the second real dataset. The first and third rows show images of the first band of the resulting images, and the second and fourth rows show false-color images of the first, second, and third bands of the resulting images.

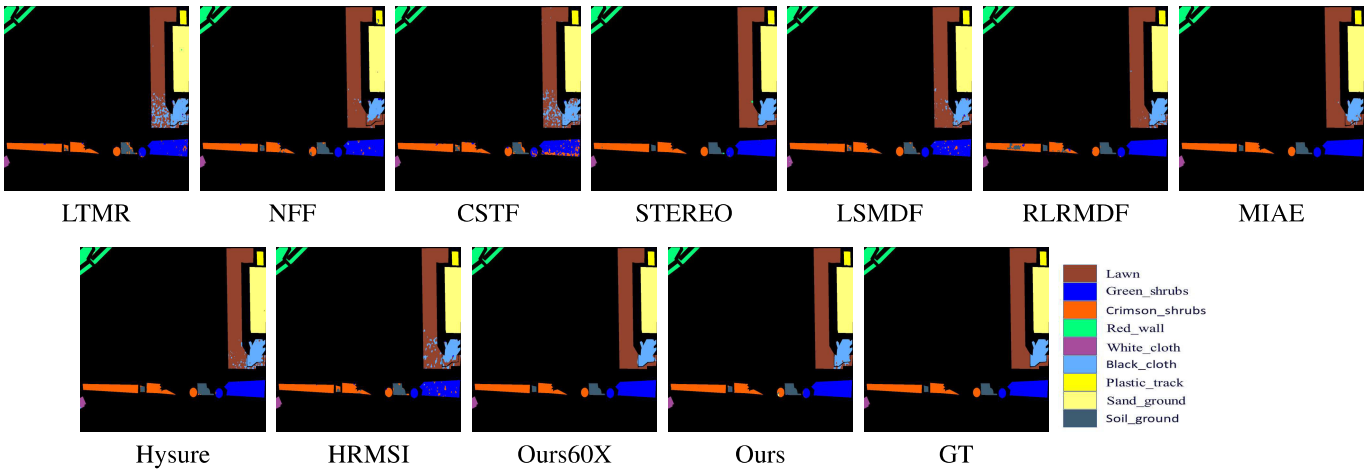


Fig. 10. Classification results of different fusion images using the SSFTT classification algorithm.

reserved 99% for testing. In the case of SSRNet, the epoch was set to 5, while the other parameters remained consistent with those specified in the original study. For HybridSN, the epoch was adjusted to 200, with all other parameters maintained by the original settings. During the preprocessing stage, when performing PCA dimensionality reduction, the PCA parameter for the fused image was set to 30, while for MSI (RGB) images, it was set to 3. Note that, in HybridSN, due to the utilization of a 3-D convolutional kernel with a parameter greater than 3 in the channel dimension, classification results for MSI images on HybridSN cannot be obtained.

Table X displays the average values of the classification metrics resulting from five classification operations using three classification algorithms on images fused by various fusion methods from our captured real dataset, as well as on MSI images. The optimal results in the table are highlighted in red, the second-best results are highlighted in blue, and the third-best results are indicated in bold. As can be seen from

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.

the table, the fused image (Ours 60X) obtained by fusing lrLR-HSI using our method gives the best classification results.

In addition, we note that the results (Ours) of fusing LR-HSI using our method achieve only the third-best classification result. This situation can be attributed to two main reasons. First, our method primarily relies on utilizing prior information related to the singular value correlation to extract spectral information. However, due to computational resource limitations, the large spatial dimensions of real datasets restrict the parameter **p** associated with spectral information from being set to larger values (setting **p** too large would lead to memory overflow during singular value decomposition computations). In contrast, the smaller spatial dimensions of LrLR-HSI result in relatively more concentrated spectral information. Therefore, when using a relatively small parameter **p**, the fusion of LrLR-HSI can extract a relatively large amount of spectral information. As a result, the fusion result Ours 60X outperforms Ours. Second, compared to the STEREO method, the visual fusion results in Fig. 9 reveal that STEREO's fusion results exhibit more prominent stripes, resulting in greater disparities between individual pixels in the fused images. Consequently, during the PCA dimensionality reduction in the preprocessing stage of the classification algorithm, STEREO retained more information, leading to improved classification results. However, when considering both subjective visual presentation and objective classification metrics, our proposed method not only showcased impressive visual effects but also demonstrated exceptional classification performance.

Of particular note is that our fusion method outperforms all comparison methods in both subjective visual effects and objective classification metrics when implementing high magnification super-resolution (Ours 60X). Fig. 10 illustrates the classification results of various fusion images using the SSFTT algorithm. The classification results also demonstrate that the images fused by our proposed method exhibit superior classification performance, thus validating the practicality of our approach.

V. Conclusion

This article proposes a novel super-resolution algorithm for the high-magnification fusion of hyperspectral and multispectral images. This algorithm is based on a coupled prior-constrained joint spatial–spectral optimization framework, which effectively enhances the spatial–spectral feature extraction capability of the algorithm. In addition, we propose a method that utilizes a fast fusion solution to initialize spectral bases and coefficient matrices, which facilitates the extraction of spatial–spectral information. Finally, we introduce an error backpropagation module that improves the fusion performance through simple operations. We compared our algorithm to 13 state-of-the-art algorithms on both simulated and real datasets. The experimental results demonstrated that our method achieved optimal fusion performance at a high-magnification super-resolution scale of up to 64×. Furthermore, our proposed method also achieved state-of-the-art performance in conventional super-resolution fusion experiments. The visualization and classification results of the experiments on two pairs of real datasets confirmed the usefulness of our method. However, it is evident from the text that the fusion results of our approach at low-magnification super-resolution scales were not significantly improved compared to those observed at high-magnification super-resolution scales. Therefore, we plan to focus on improving the fusion performance at low-magnification super-resolution scales in future research efforts.

References

[1] F. D. van der Meer et al., "Multi- and hyperspectral geologic remote sensing: A review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 14, no. 1, pp. 112–128, Feb. 2012.

[2] Y. Cui, B. Zhang, W. Yang, X. Yi, and Y. Tang, "Deep CNN-based visual target tracking system relying on monocular image sensing," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2018, pp. 1–8.

[3] Y. Cui et al., "End-to-end visual target tracking in multi-robot systems based on deep convolutional neural network," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops*, Sep. 2017, pp. 1113–1121.

[4] W. K. Wong, Z. Lai, Y. Xu, J. Wen, and C. P. Ho, "Joint tensor feature analysis for visual object recognition," *IEEE Trans. Cybern.*, vol. 45, no. 11, pp. 2425–2436, Nov. 2015.

[5] R. Hang, Q. Liu, and Z. Li, "Spectral super-resolution network guided by intrinsic properties of hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 30, pp. 7256–7265, 2021.

[6] S. Zhou et al., "Hyperspectral imaging of beet seed germination prediction," *Infr. Phys. Technol.*, vol. 108, Aug. 2020, Art. no. 103363.

[7] A. Hirai, "Mineral discrimination by combination of multispectral image and surrounding hyperspectral image," *J. Appl. Remote Sens.*, vol. 13, no. 2, May 2019, Art. no. 024517.

[8] J. Yang, Y. Zhao, J. C. Chan, and S. G. Kong, "Coupled sparse denoising and unmixing with low-rank constraint for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1818–1833, Mar. 2016.

[9] Y. Li, H. Zhu, and S. Yu, "High-magnification super-resolution reconstruction of image with multi-task learning," *Electronics*, vol. 11, no. 9, p. 1412, Apr. 2022.

[10] R. Dian and S. Li, "Hyperspectral image super-resolution via subspace-based low tensor multi-rank regularization," *IEEE Trans. Image Process.*, vol. 28, no. 10, pp. 5135–5146, Oct. 2019.

[11] J. Long, Y. Peng, J. Li, L. Zhang, and Y. Xu, "Hyperspectral image super-resolution via subspace-based fast low tensor multi-rank regularization," *Infr. Phys. Technol.*, vol. 116, Aug. 2021, Art. no. 103631.

[12] J. Bieniarz, R. Müller, X. X. Zhu, and P. Reinartz, "Hyperspectral image resolution enhancement based on joint sparsity spectral unmixing," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 2645–2648.

[13] Q. Xie et al., "Multispectral images denoising by intrinsic tensor sparsity regularization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 1692–1700.

[14] R. Dian, S. Li, L. Fang, and Q. Wei, "Multispectral and hyperspectral image fusion with spatial-spectral sparse representation," *Inf. Fusion*, vol. 49, pp. 262–270, Sep. 2019.

[15] J. Peng et al., "Low-rank and sparse representation for hyperspectral image processing: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 10–43, Mar. 2022.

[16] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[17] J. Long, Y. Peng, L. Zhao, T. Zhou, and J. Li, "Semi-blind hyperspectral and multispectral image fusion based on a non-factorization model," *Infr. Phys. Technol.*, vol. 125, Sep. 2022, Art. no. 104232.

[18] M. Simoes, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[19] X. Zhang, W. Huang, Q. Wang, and X. Li, "SSR-NET: Spatial–spectral reconstruction network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5953–5965, Jul. 2021.

[20] Q. Xie, M. Zhou, Q. Zhao, Z. Xu, and D. Meng, "MHF-Net: An interpretable deep network for multispectral and hyperspectral image fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 3, pp. 1457–1473, Mar. 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.

[21] J. Yao, D. Hong, J. Chanussot, D. Meng, X. Zhu, and Z. Xu, "Cross-attention in coupled unmixing nets for unsupervised hyperspectral super-resolution," in *Proc. Eur. Conf. Comput. Vis.* Glasgow, U.K.: Springer, Aug. 2020, pp. 208–224.

[22] R. Dian, S. Li, and X. Kang, "Regularizing hyperspectral and multispectral image fusion by CNN denoiser," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 3, pp. 1124–1135, Mar. 2021.

[23] W. Dong, C. Zhou, F. Wu, J. Wu, G. Shi, and X. Li, "Model-guided deep hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 5754–5768, 2021.

[24] J. Liu, Z. Wu, L. Xiao, and X.-J. Wu, "Model inspired autoencoder for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022, Art. no. 5522412, doi: 10.1109/TGRS.2022.3143156.

[25] R. C. Hardie, M. T. Eismann, and G. L. Wilson, "MAP estimation for hyperspectral image resolution enhancement using an auxiliary sensor," *IEEE Trans. Image Process.*, vol. 13, no. 9, pp. 1174–1184, Sep. 2004.

[26] Y. Zhang, S. De Backer, and P. Scheunders, "Noise-resistant wavelet-based Bayesian fusion of multispectral and hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 11, pp. 3834–3843, Nov. 2009.

[27] Y. Zhang, A. Duijster, and P. Scheunders, "A Bayesian restoration approach for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 9, pp. 3453–3462, Sep. 2012.

[28] N. Akhtar, F. Shafait, and A. Mian, "Bayesian sparse representation for hyperspectral image super resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3631–3640.

[29] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[30] H. Song, B. Huang, K. Zhang, and H. Zhang, "Spatio-spectral fusion of satellite images based on dictionary-pair learning," *Inf. Fusion*, vol. 18, pp. 148–160, Jul. 2014.

[31] X. Bresson and T. F. Chan, "Fast dual minimization of the vectorial total variation norm and applications to color image processing," *Inverse Problems Imag.*, vol. 2, no. 4, pp. 455–484, 2008.

[32] R. Kawakami, Y. Matsushita, J. Wright, M. Ben-Ezra, Y.-W. Tai, and K. Ikeuchi, "High-resolution hyperspectral imaging via matrix factorization," in *Proc. CVPR*, Jun. 2011, pp. 2329–2336.

[33] W. Dong et al., "Hyperspectral image super-resolution via non-negative structured sparse representation," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2337–2352, May 2016.

[34] N. Akhtar, F. Shafait, and A. Mian, "Sparse spatio-spectral representation for hyperspectral image super-resolution," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2014, pp. 63–78.

[35] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.

[36] R. Dian, L. Fang, and S. Li, "Hyperspectral image super-resolution via non-local sparse tensor factorization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3862–3871.

[37] C. I. Kanatsoulis, X. Fu, N. D. Sidiropoulos, and W.-K. Ma, "Hyperspectral super-resolution: A coupled tensor factorization approach," *IEEE Trans. Signal Process.*, vol. 66, no. 24, pp. 6503–6517, Dec. 2018.

[38] R. Dian, S. Li, A. Guo, and L. Fang, "Deep hyperspectral image sharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 11, pp. 5345–5355, Nov. 2018.

[39] X. Han, J. Yu, J. Luo, and W. Sun, "Hyperspectral and multispectral image fusion using cluster-based multi-branch BP neural networks," *Remote Sens.*, vol. 11, no. 10, p. 1173, May 2019.

[40] W. Sun et al., "MLR-DBPFN: A multi-scale low rank deep back projection fusion network for anti-noise hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, Art. no. 5522914, doi: 10.1109/TGRS.2022.3146296.

[41] K. Ren, W. Sun, X. Meng, G. Yang, J. Peng, and J. Huang, "A locally optimized model for hyperspectral and multispectral images fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022, Art. no. 5519015, doi: 10.1109/TGRS.2021.3133670.

[42] F. Zhou, R. Hang, Q. Liu, and X. Yuan, "Pyramid fully convolutional network for hyperspectral and multispectral image fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 5, pp. 1549–1558, May 2019.

[43] J. Liu, Z. Wu, L. Xiao, J. Sun, and H. Yan, "A truncated matrix decomposition for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 29, pp. 8028–8042, 2020.

[44] X. Han, J. Yu, J.-H. Xue, and W. Sun, "Hyperspectral and multispectral image fusion using optimized twin dictionaries," *IEEE Trans. Image Process.*, vol. 29, pp. 4709–4720, 2020.

[45] J. Long and Y. Peng, "Blind fusion of hyperspectral multispectral images based on matrix factorization," *Remote Sens.*, vol. 13, no. 21, p. 4219, Oct. 2021.

[46] D. Arthur and S. Vassilvitskii, "K-means++: The advantages of careful seeding," vol. 8, in *Proc. 18th Annu. ACM-SIAM Symp. Discrete Algorithms*, Jan. 2007, pp. 1027–1035.

[47] F. Dell'Acqua, P. Gamba, A. Ferrari, J. A. Palmason, J. A. Benediktsson, and K. Árnason, "Exploiting spectral and spatial information in hyperspectral urban data with high resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 1, no. 4, pp. 322–326, Oct. 2004.

[48] A. Chakrabarti and T. Zickler, "Statistics of real-world hyperspectral images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2011, pp. 193–200.

[49] L. Wald, "Quality of high resolution synthesised images: Is there a simple criterion," in *Proc. 3rd Conf. 'Fusion Earth Data, Merging Point Meas., Raster Maps Remotely Sensed Images'*, Sophia Antipolis, France, Jan. 2000, pp. 99–103.

[50] Q. Wei, J. Bioucas-Dias, N. Dobigeon, and J.-Y. Tourneret, "Hyperspectral and multispectral image fusion based on a sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3658–3668, Jul. 2015.

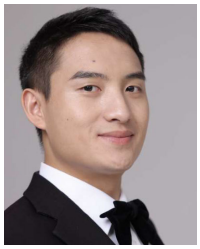
[51] N. Yokoya, C. Grohnfeldt, and J. Chanussot, "Hyperspectral and multispectral data fusion: A comparative review of the recent literature," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 2, pp. 29–56, Jun. 2017.

[52] R. Dian, S. Li, and L. Fang, "Learning a low tensor-train rank representation for hyperspectral image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2672–2683, Sep. 2019.

[53] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[54] D. Song, C. Yang, B. Wang, J. Zhang, H. Gao, and Y. Tang, "SSRNet: A lightweight successive spatial rectified network with noncentral positional sampling strategy for hyperspectral images classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–15, 2023, Art. no. 5519115, doi: 10.1109/TGRS.2023.3301310.

[55] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.



Jian Long received the B.S. and M.S. degrees in computer science from the National University of Defense Technology, Changsha, China, in 2018 and 2021, respectively, where he is currently pursuing the Ph.D. degree in computer science.



Yibing Zhan (Member, IEEE) received the bachelor's and doctor's degrees from the School of Information Science and Technology, University of Science and Technology of China, Hefei, China, in 2012 and 2018, respectively.

After graduating with a doctor's degree, he served as an Associate Researcher with the School of Computer Science, Hangzhou Dianzi University, Hangzhou, China, from 2018 to 2020. Now, he works with the JD Explore Academy, Beijing, China, as an Algorithm Scientist. He has published many scientific papers in top conferences and journals, such as NeurIPS, CVPR, ACM MM, ICCV, IJCV, IEEE TRANSACTIONS ON IMAGE PROCESSING, and IEEE TRANSACTIONS ON MULTIMEDIA. He mainly explores multimodal learning, large-scale models, and graph neural networks.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.



Zhengbin Pang received the B.S., M.S., and Ph.D. degrees in computer science from the National University of Defense Technology (NUDT), Changsha, China, in 1994, 1997, and 2007, respectively.

He is currently a Professor with the School of Computer, NUDT. His research interests include parallel and distributed computing and high-performance computer systems.



Tong Zhou received the Ph.D. degree in electronic science and technology from the National University of Defense Technology, Changsha, China, in 2019.

He is currently an Associate Professor with the College of Advanced Interdisciplinary Studies, National University of Defense Technology. His research interests include topological materials and their growth and hyperspectral image processing.



Xueqiong Li received the B.S. degree in communication and information engineering from the College of Transportation, Southeast University, Nanjing, China, in 2013, and the M.S. and Ph.D. degrees in communication and information engineering from the College of Electronic Science and Engineering, National University of Defense Technology, Changsha, China, in 2015 and 2020, respectively.

She is currently an Associate Researcher with the School of Computer Science, National University of Defense Technology. Her research interests include multimodal information processing.



Long Lan received the Ph.D. degree in computer science from the National University of Defense Technology, Changsha, China, in 2017.

He was a Visiting Ph.D. Student at the University of Technology Sydney, Ultimo, NSW, Australia, from 2015 to 2017. He is currently an Associate Professor with the College of Computer Science and Technology, National University of Defense Technology. He has published more than 40 journal articles and conference papers, such as IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IJCV, IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON MULTIMEDIA, IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, NeurIPS, ICLR, MM, KDD, AAAI, IJCAI, and ECCV. His research interests include computer vision and machine learning, especially multiobject tracking, transfer learning, and few-shot learning.

Dr. Lan served as a Senior Program Committee Member of IJCAI'21.



Yuanxi Peng was born in 1966. He received the B.S. degree in computer science from Sichuan University, Chengdu, China, in 1988, and the M.S. and Ph.D. degrees in computer science from the National University of Defense and Technology (NUDT), Changsha, China, in 1998 and 2001, respectively.

He was a Visiting Professor with the Department of Electronic and Computer Engineering, University of Toronto, Toronto, ON, Canada, from 2010 to 2011. He has been a Professor at the Computer School, NUDT, since 2011. His research interests are in the areas of hyperspectral image processing, high-performance computing, multi and many-core architectures, on-chip networks, cache coherence protocols, and architectural support for parallel programming.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:51:31 UTC from IEEE Xplore. Restrictions apply.